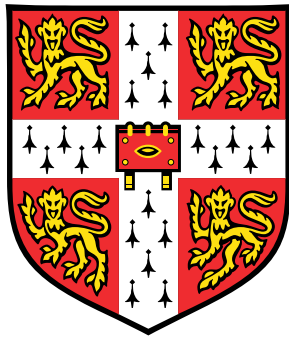


Improving Breast Cancer Screening with CNN-Wavelet Scattering



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I would like to dedicate this thesis to my parents.

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements.

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Abstract

Breast-cancer screening relies on mammography, yet the public datasets available for building computer-aided diagnosis systems are small, so convolutional neural networks (CNNs) trained on them tend to overfit; handcrafted, training-free texture features such as the wavelet scattering transform (WST) offer a stable complement. This dissertation presents a Python replication of the CNN–wavelet-scattering fusion framework of Razali et al. on the INbreast dataset, evaluated on two distinct tasks: fatty-versus-fibroglandular tissue-density classification and benign-versus-malignant mass classification. The pipeline comprises percentile-clip normalisation, ground-truth pectoral-muscle removal and CLAHE preprocessing; strictly image-level (group-aware) data splitting to prevent patch-level leakage; wavelet-scattering and fine-tuned ResNet18 feature branches; and their feature-level fusion. Beyond replication, the wavelet-scattering branch is extended along two threads: a scattering-covariance descriptor for tissue, and — for mass — a representation that preserves the scattering spatial map at a lower invariance scale and learns over it with a small CNN head, with a lesion-relative region-aware decomposition as its interpretable instantiation and a late-fusion diagnostic. All headline results use a repeated five-seed, AUROC-led protocol. The tissue CNN and fusion models are near-saturated, and scattering covariance is the strongest tissue descriptor (AUROC ≈ 0.99); on mass, paper-style fusion improves on either branch alone, scattering covariance gives no gain, while preserving the scattering map and learning over it lifts mass AUROC from ≈ 0.71 to ≈ 0.92 (whole-patch map-CNN ≈ 0.92 ; the region-aware instantiation ≈ 0.91 , on par within seed noise), and a decision-level late fusion is numerically strongest (≈ 0.94) but is reported as an ensemble rather than the core contribution. The central finding is that, for tissue texture, richer *scalar* scattering descriptors suffice, whereas for mass lesions *preserving the spatial structure* of the representation matters more than changing scalar features or classifiers — though, on the small mass set (~ 24 held-out patches), all mass estimates carry wide across-seed bands.

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Chapter 1

Introduction

Breast cancer is the most prevalent malignancy among women worldwide and the most frequently diagnosed cancer globally, with approximately 2.3 million new cases each year [1]. Early detection through routine mammographic screening substantially improves patient outcomes, yet the interpretation of mammograms remains a challenging and time-consuming task for radiologists. Computer-aided diagnosis (CAD) systems that leverage computer vision have therefore become an active area of research, offering the potential to assist clinicians in detecting and characterising suspicious findings.

Modern CAD approaches overwhelmingly rely on convolutional neural networks (CNNs), which have demonstrated strong performance for a range of medical imaging tasks [2]. However, two significant limitations persist. First, CNNs are data-hungry: their supervised training paradigm requires large volumes of annotated data, while publicly available mammography datasets remain relatively small. Second, conventional CNN architectures primarily extract spatial-domain features and may neglect complementary information present in the frequency domain, such as the multi-scale textural patterns that characterise different tissue types.

The wavelet scattering transform (WST) provides an attractive solution to both issues. Introduced by Mallat [3] and further developed by Bruna and Mallat [4], scattering transforms cascade the input through banks of predefined wavelet filters and non-linear modulus operations to produce feature representations that are locally invariant to translations and stable to small deformations. Crucially, the WST requires no training data at all — it is a fixed mathematical operation — and it captures rich multi-scale, multi-orientation texture information in the frequency domain.

Razali et al. [5] proposed a hybrid framework that fuses CNN features with WST features (and additionally GLCM texture descriptors [6]) for classifying breast tissue in mammograms, demonstrating that this combination outperforms either branch alone. Their method was

evaluated on the INbreast dataset [7] for two tasks: distinguishing fatty from fibroglandular background tissue, and classifying mass lesions as benign or malignant.

This project aims to replicate the methodology of Razali et al. in Python and to reproduce the classification results reported therein. The pipeline is implemented in a modular fashion so that each component — preprocessing, patch extraction, wavelet scattering features, CNN features, and the fusion classifier — can be evaluated independently. The work proceeded iteratively, beginning with the wavelet scattering branch on the fatty versus fibroglandular tissue task, extending to the CNN branch on the same task, then repeating both branches on the benign versus malignant mass task, and finally implementing the CNN + WS feature fusion that represents the paper’s headline contribution. The development was not linear: an early CNN run revealed a data-leakage issue that required revisiting the train–test split methodology for both branches, which is documented as part of the development narrative in Section 4. This report compares all three model configurations (WS-only, CNN-only, CNN + WS) against the paper’s reported Table 6 (tissue) and Table 7 (mass) benchmarks. The GLCM branch reported in the paper (Models 3, 4, 6 of their Table 6) was excluded from the scope of this project on supervisor advice, since the paper itself shows GLCM contributes negligible improvement when combined with the WS branch.

Beyond replication, the project extends the wavelet-scattering branch along two threads. For the *tissue/background* texture task it adopts the *scattering covariance* representation (Cheng & Ménard [8]; supervisor-suggested), which augments the standard scattering coefficients with cross-scale and cross-orientation correlation statistics while remaining a compact scalar descriptor. For the *mass* task — where paper-style *averaged* scattering is weak because spatial averaging discards *where* the lesion structure lies — it instead *preserves* the scattering spatial map at a lower invariance scale ($J = 3$) and learns over it with a small WS-map CNN head; a *region-aware* decomposition (lesion core, margin, whole-lesion, and surrounding-context regions) is the interpretable, morphology-conditioned instantiation of this idea. The contribution is not merely higher accuracy but a methodological finding: *scalar* covariance statistics are effective for tissue texture, whereas mass lesions require *spatial preservation* of the scattering map (the region decomposition being a principled, interpretable refinement that performs on par rather than the source of the gain) rather than a change of scalar feature or classifier. A decision-level late fusion of the WS-map CNN with a frozen-CNN comparator is the numerically strongest mass configuration, but it is reported as an ensemble result rather than the core contribution.

1.1 Project Goals

The specific objectives of this project, following the brief set out by the project supervisor, are:

1. Load and explore the INbreast clinical dataset (DICOM mammographic images).
2. Reproduce the primary pipeline from the reference publication in a modular fashion, enabling comparison of results with and without the wavelet scattering transform.
3. Reproduce the classification results — including performance metrics and model comparisons — both (a) with and (b) without the scattering wavelet transform.
4. Investigate wavelet-scattering extensions for the two tasks, including scattering covariance for tissue/background texture and region-aware scattering-map preservation for mass lesions.

1.2 Report Structure

Section 2 reviews the theoretical background to the project — mammography and breast cancer screening, handcrafted texture features, wavelets, the wavelet scattering transform and its use for classification, convolutional neural networks, and feature fusion. Section 3 details the dataset, preprocessing pipeline, patch extraction (background tissue and mass ROIs), wavelet scattering and CNN feature extraction branches, and classifier design. Section 4 presents the paper-style replication results as a chronological development narrative, covering each stage in the order it was carried out and comparing the outcomes against the paper’s benchmarks. Section 5 reports the extension experiments: a screening of supporting non-region-aware methods, the scattering covariance representation for the tissue task, and the region-aware wavelet-scattering deep dive for the mass task. Section 6 discusses sources of discrepancy, lessons learned, the data-quality issues uncovered, limitations, and potential improvements. Section 7 concludes the report with a summary and directions for further work.

Chapter 2

Background

This section sets out the theory the project rests on. It starts with the clinical motivation (mammographic screening and the two classification tasks), then builds up the feature-extraction side: why handcrafted texture features are useful, what wavelets are, and how the wavelet scattering transform turns wavelets into a stable feature representation. It then covers, more briefly, convolutional neural networks and the idea of fusing the two feature types, since both are part of the wider pipeline this project replicates. The aim is to explain the ideas, not to re-derive them; precise implementation choices and parameter values are given later in Section 3.

2.1 Mammography and breast cancer screening

Breast cancer is the most commonly diagnosed cancer in women worldwide [1], and X-ray mammography is the primary imaging tool used to screen for it. A mammogram is a low-dose X-ray projection of the breast; the goal of screening is to find suspicious changes early, while they are still small and more treatable. Large-scale reviews of screening programmes estimate that inviting women for regular mammographic screening reduces breast cancer mortality by roughly 20%, albeit at the cost of some overdiagnosis [9]. In practice a radiologist reads each mammogram and records findings using a standardised vocabulary — the American College of Radiology’s BI-RADS lexicon, which encodes both an overall suspicion category and a breast-density category [10].

Reading mammograms is time-consuming and subject to inter-reader variability, which has motivated a long line of computer-aided diagnosis (CAD) research that tries to support the radiologist with automated image analysis [2]. This project, following Razali et al. [5], looks at two distinct classification problems that arise in this setting. The first is **tissue-density classification**: deciding whether a region of breast tissue is fatty or fibroglandular

(dense). Density matters clinically because dense tissue both raises cancer risk and makes tumours harder to see [11], and it is one of the BI-RADS categories [10]. The second is **mass classification**: deciding whether a detected mass lesion is benign or malignant. These two tasks are different in character — density is a property of relatively uniform background texture, whereas a mass is a localised structured object — and this distinction recurs throughout the report.

2.2 Feature extraction and texture in mammograms

Before any classifier can be trained, an image (or image patch) has to be turned into a numerical feature vector. Broadly there are two families of approach. *Handcrafted* features are computed with a fixed, human-designed procedure; *learned* features are optimised from data by a neural network (Section 2.6). Handcrafted features have a particular appeal in medical imaging because they need no training data, are deterministic, and are easier to reason about — useful when annotated datasets are small, as they typically are for mammography.

For mammograms, much of the discriminative information lives in *texture*: the spatial pattern of intensity variation rather than the overall brightness. Fatty and fibroglandular tissue differ mainly in how their intensity fluctuates across the image, and benign and malignant masses differ in the regularity and sharpness of their internal structure and margins. A classical way to quantify texture is the grey-level co-occurrence matrix (GLCM) [6], which summarises how often pairs of intensities occur at given offsets. A limitation of single-scale texture descriptors is that breast structures appear at many sizes at once — from fine trabecular patterns to larger glandular regions — so a representation that captures structure at *multiple scales* is better matched to the problem. This multi-scale requirement is what motivates wavelet-based features.

2.3 Wavelets and multi-scale analysis

A wavelet is a small, localised oscillating waveform. By dilating (stretching) and rotating a single mother wavelet, one obtains a family of filters tuned to different scales and orientations [12]. This multi-scale (multiresolution) decomposition — representing a signal as a coarse approximation plus added detail at successively finer scales — is the foundation of wavelet image analysis [13]. Convolution with such a filter bank measures how much “edge-like” or “oscillatory” content the image has at each scale and orientation, and at each location. This is the basic idea of multi-scale image analysis: coarse wavelets respond to large, smooth structures, while fine wavelets respond to small, sharp ones.

Wavelets are well suited to texture and medical image analysis for two reasons. First, because they are localised in space (unlike the global sinusoids of the Fourier transform), they are stable under small geometric deformations — a slightly warped texture produces only a slightly changed wavelet response [3]. Second, the multi-scale, multi-orientation decomposition naturally matches the way biological texture is organised across scales. The drawback is that a raw wavelet transform is *not* translation invariant: shifting the image shifts all the coefficients, so two patches of the same tissue type at different positions look different to a classifier. Removing this nuisance variability, while keeping the useful texture information, is exactly what the scattering transform is designed to do.

2.4 The wavelet scattering transform

The wavelet scattering transform (WST), introduced by Mallat [3] and Bruna and Mallat [4], builds a feature representation that is locally invariant to translation and stable to small deformations, while still preserving high-frequency texture information. It does this by cascading three simple operations:

1. **Wavelet convolution.** The signal x is convolved with an oriented wavelet ψ_λ (indexed by a scale and orientation λ), giving $x \star \psi_\lambda$. This is the same multi-scale filtering described above.
2. **Modulus non-linearity.** The complex wavelet response is passed through the modulus $|x \star \psi_\lambda|$. Taking the modulus removes the oscillatory phase and produces a smooth, non-negative “envelope” of the wavelet response. A non-linearity is essential: without it, the subsequent averaging step would simply destroy the information, because wavelets integrate to zero [4].
3. **Local averaging.** The envelope is averaged against a low-pass filter ϕ_J at scale 2^J , i.e. $|x \star \psi_\lambda| \star \phi_J$. This averaging is what creates local translation invariance: shifts smaller than the window 2^J leave the averaged value almost unchanged.

Averaging discards the finer detail of the envelope, but that lost detail is not thrown away — it is recovered by applying the wavelet–modulus step *again* to the envelope before averaging, giving a second “layer”. Iterating in this way defines the **scattering orders**. The zeroth order is just the smoothed image $S_0x = x \star \phi_J$; the first order is $S_1x = |x \star \psi_{\lambda_1}| \star \phi_J$; the second order is $S_2x = ||x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}| \star \phi_J$; and so on. Concatenating the coefficients of all orders gives the scattering representation. (The precise definitions and parameters used in this project are given in Section 3.4.)

Two theoretical properties make this representation attractive. **Translation stability:** the averaging by ϕ_J makes the descriptor invariant to translations small relative to the scale 2^J , so the same tissue at different positions maps to nearly the same features [3]. **Deformation stability:** because wavelets are localised and the modulus is a smooth non-linearity, a small deformation of the image produces only a small, bounded change in the scattering coefficients (a Lipschitz-continuity property) [3, 4]. In practice this means the WST is robust to exactly the kinds of small geometric variation seen between mammogram patches of the same class. Bruna and Mallat further show that most of the representation’s energy is captured within the first two orders, with negligible energy beyond order two, which is why second-order scattering is the standard choice [4]. Crucially, the wavelet filters are fixed in advance — the WST has no parameters to learn — so it can be applied directly to small datasets.

2.5 Scattering features for classification

The scattering transform outputs coefficient *maps*: for each path (the chain of scales and orientations defining $S_0, S_1, S_2, \dots$) there is a small spatially-subsampled image of averaged values. To obtain a fixed-length feature vector suitable for a standard classifier, these maps are reduced to scalars — most simply by *spatially averaging* each map, giving one number per path. Stacking those numbers produces the feature vector that is fed to a classifier. This is the representation used in the original scattering papers [4] and in the reference paper for this project [5].

Once an image is represented as such a vector, any general-purpose classifier can be used, because the hard work of building an informative, invariant representation has already been done by the transform. Bruna and Mallat themselves classify scattering vectors with a generative PCA-based classifier and with a kernel support vector machine (SVM), and observe that simpler, lower-capacity classifiers tend to do better when training samples are scarce, with the more flexible SVM only overtaking them as the dataset grows [4]. Other standard choices include logistic regression and nearest-neighbour methods [14]; the reference paper uses a Subspace k -nearest-neighbours ensemble, which combines many k -NN classifiers each trained on a random subset of features [15, 14, 16]. The key point is that the classifier sits *on top of* the fixed scattering features and can be chosen independently of them.

How the coefficient maps are summarised is itself a design choice with several options. *Spatial averaging* (one scalar per path) is the most invariant and compact, and is appropriate when the patch is treated as a homogeneous texture. *Preserving the coefficient maps* (keeping the spatial grid, or richer per-path statistics) retains information about *where* structure occurs,

which can matter for localised objects but reduces translation invariance. A third option is the *scattering covariance* of Cheng and Ménard [8], which augments the averaged coefficients with correlation statistics computed *between* scattering paths. This captures higher-order dependencies in the texture — information that single-path averages miss — while still producing a compact set of scalar descriptors rather than a flattened spatial tensor. This project uses spatial averaging for its baseline scattering features and explores the scattering covariance as a richer alternative (Sections 3.4 and 5.2).

2.6 Convolutional neural networks

A convolutional neural network (CNN) is the learned counterpart to the handcrafted pipeline above [17]. Like the WST, a CNN applies banks of convolutional filters across the image, but its filters are *learned* from labelled data by gradient descent rather than fixed in advance. Stacking many convolutional layers, interleaved with non-linearities and *pooling* operations that progressively reduce spatial resolution, lets early layers respond to simple patterns (edges, blobs) and later layers to increasingly complex, task-specific structures. A final set of fully connected layers maps the resulting features to class scores. Deep CNNs such as the residual network (ResNet) architecture [18] achieve strong results across medical imaging tasks [2].

The trade-off relative to the scattering transform is essentially data versus flexibility. A CNN can, in principle, learn whatever features best separate the classes, but it needs a large amount of labelled data to do so and can overfit when data is scarce — a real concern for mammography datasets, which are small. The WST, by contrast, has no learnable filters: it cannot adapt to the data, but it also cannot overfit, and it works directly on small datasets. The two approaches are therefore complementary, which is the motivation for combining them.

2.7 Fusion of scattering and CNN features

Fusion means combining the two feature types into a single classification decision, in the hope of getting the best of both: the data-driven flexibility of the CNN and the stable, training-free texture description of the scattering transform. There are two broad strategies. In **feature-level fusion**, the feature vectors from each branch are concatenated into one longer vector that a single classifier is trained on. In **decision- (or model-) level fusion**, each branch produces its own prediction and these are combined afterwards, for example by averaging probabilities or majority voting. The reference paper performs feature-level

fusion, concatenating CNN and scattering features before the final classifier, and reports that the fused representation outperforms either branch alone [5]. The intuition is that the scattering features supply robust texture information that helps most precisely when the CNN is data-starved and at risk of overfitting. Ensemble and combination methods more generally are a well-established way to improve classifier robustness [15].

2.8 Why spatial information matters for mass lesions

Wavelet scattering buys its stability through *spatial averaging*: each coefficient map $S_n(\vec{x})$ is collapsed to a single scalar $\langle S_n \rangle$ per path, which makes the descriptor invariant to small translations and deformations. For a roughly homogeneous texture — such as a background-tissue patch in the fatty-versus-fibroglandular task — this is exactly the right inductive bias: the discriminative signal is the *statistics* of the texture, not where any particular structure sits, so averaging discards only nuisance position information.

A mass lesion is different. Malignancy is encoded less in a single homogeneous texture and more in the *spatial arrangement* of distinct sub-regions: a dense lesion *core*, the *margin* or boundary (spiculated or irregular margins are a classical malignancy cue, whereas smooth circumscribed margins suggest benignity), and the immediately *surrounding* parenchyma (architectural distortion, tethering). Spatially averaging the scattering maps over the whole patch blends these regions together and erases precisely the boundary-versus-core contrast that separates benign from malignant masses. This is consistent with the empirical pattern reported later (Chapter 4): averaged wavelet scattering is a strong tissue descriptor but a weak mass descriptor.

This observation motivates the project’s main mass-side extension. Rather than discard *where* the lesion structure lies, we (i) keep the scattering *maps* at a lower invariance scale ($J = 3$, which retains a 28×28 spatial grid per coefficient channel) and let a small convolutional head learn over them, and (ii) explicitly decompose each lesion into core/inner, boundary, and surrounding-context regions before scattering, so the descriptor is conditioned on lesion morphology (Chapter 5). Tissue-density classification, being texture-like, does not need this spatial conditioning and is instead improved on the WS side by a richer *scalar* descriptor, the scattering covariance.

2.9 Relation to this project

The pipeline replicated and extended in this project follows directly from the ideas above. Mammograms are first preprocessed (normalisation, pectoral-muscle removal, breast mask-

ing, contrast enhancement) so that later feature extraction sees clean, comparable inputs. Fixed-size patches are then extracted for the two tasks: background tissue patches for fatty-versus-fibroglandular density classification, and mass-lesion patches for benign-versus-malignant classification. Each patch is passed through the wavelet scattering transform (and, as a supervisor-suggested extension, the scattering covariance) to produce handcrafted texture features, and separately through a fine-tuned CNN to produce learned features. The two feature types are evaluated on their own and then fused, with a classifier trained on top to make the final tissue or mass prediction. The remainder of the report makes each of these steps concrete (Section 3) and reports how well they reproduce the reference paper’s results (Section 4).

Chapter 3

Data, Preprocessing, Methods, and Experimental Protocol

This section describes the end-to-end pipeline: from loading and preprocessing the raw mammographic data, through patch extraction and tissue labelling, to wavelet scattering feature computation and classification. The pipeline is implemented across several Jupyter notebooks, each corresponding to one stage, and the full codebase is available in the accompanying project repository.

3.1 Dataset: INbreast

We use the INbreast dataset [7], a publicly available full-field digital mammography (FFDM) database comprising 410 DICOM images acquired at a breast centre in Porto, Portugal. Each case includes standard cranio-caudal (CC) and medio-lateral oblique (MLO) views.

Three complementary metadata sources are linked by a common numeric `file_id`:

- **INbreast.xls** — per-image metadata including BI-RADS assessment category [10], ACR breast density score (1–4), laterality, and acquisition view.
- **OsiriX XML annotations** — plist-format files (343 of 410 images) containing region-of-interest (ROI) polygons for lesions such as masses.
- **Pectoral muscle masks** — LabelMe-format JSON polygon annotations [19] covering 202 MLO-view images, providing ground-truth segmentations for the pectoral muscle.

For the tissue classification task, a binary density label is derived from the ACR score: ACR 1–2 maps to non-dense (class 0), and ACR 3–4 maps to dense (class 1). For mass

classification, the BI-RADS assessment determines the label: BI-RADS 2–3 is labelled benign, and BI-RADS 4–6 is labelled malignant; BI-RADS 1 (normal) is excluded.

This binary mapping yields **409 of the 410 DICOMs** as usable for the tissue task. One row drops out at ACR cleanup: `file_id 53582540`, a BIRADS-1 normal right-MLO image whose ACR field in `INbreast.xls` is a whitespace-only string (a data-entry artefact in the original spreadsheet) and therefore cannot be placed on the binary density scale. The dropped image carries no mass annotation, no lesion ROI, and no OsiriX XML file, so the exclusion has no downstream consequence for the mass classification task or any other stage. This step and the identity of the dropped row are recorded in *Step 0.2 — Map to XLS*, §7.1.

The resulting tissue-task class distribution is moderately imbalanced: **282 non-dense** (ACR 1: 136; ACR 2: 146) vs. **127 dense** (ACR 3: 99; ACR 4: 28), i.e. approximately a **69 %/31 %** split. The majority-class baseline against which any tissue classifier should be compared is therefore around **69 %**: a model that always predicts non-dense would achieve this. Read in that light, the WS-only baseline of $\sim 84\%$ test accuracy in Section 4.1 sits roughly **15 pp above** the majority baseline, and the adapted scattering covariance result of $\sim 95.5\%$ test (Section 5.2) sits roughly **26 pp above** it. Downstream classifiers do not apply class-weight balancing; metrics are reported on the natural distribution.

3.2 Preprocessing

Each DICOM image passes through a four-stage preprocessing pipeline, implemented in *Step 1.1 — Preprocessing*:

1. **Normalisation.** The pixel array is read using `pydicom` [20] and rescaled to the interval $[0, 1]$. The reference paper converts each DICOM to PNG (an implicit min-max windowing into 8-bit) and then normalises to $[0, 1]$; we approximate this by clipping each image at its 1st and 99th percentiles and then rescaling. A direct min-max stretch was tested as an ablation and was found to compress the breast-tissue dynamic range into a narrow band near zero, because a small number of pixels per image (calibration markers, edge artefacts, sensor hot spots) sit far above the bulk distribution of breast tissue and dominate the rescaling. The percentile clip is robust to those extreme values, keeps the meaningful tissue values well-spread across $[0, 1]$, and improved 10-fold CV accuracy on the fatty-vs-fibroglandular task from $\sim 80\%$ to $\sim 83\%$ on this dataset. This is a small implementation adaptation, not a methodological change. A header audit of all 410 INbreast DICOMs (Step 0.1) further confirmed that every file uses `MONOCHROME2` photometric interpretation and that none carry `VOI LUT` or `WindowCenter/WindowWidth` metadata. Earlier versions of the loader applied a `VOI`

LUT and inverted MONOCHROME1 images defensively; both branches were no-ops on this dataset and have now been removed.

2. **Pectoral muscle removal (MLO/ML views only).** For oblique-view images with a matching ground-truth JSON mask, the pectoral muscle region is zeroed out. This prevents the pectoral muscle from being mistaken for dense breast tissue during subsequent analysis. Two independent pectoral-mask sources exist for INbreast (the LabelMe JSON files used here, and an Apple-plist XML set from the original INESC Porto release); the choice has only a small downstream effect, which is audited in Section 6.8. Coverage of the LabelMe set is **202 of the ~ 206 MLO/ML views**: 4 MLO/ML DICOMs have no pectoral mask in this source and silently bypass the pectoral step (any small pectoral region they contain is then handled by the Otsu breast-mask step below), and 1 CC view (`file_id` 50994273) carries an anomalous small polygon that the loader’s MLO/ML view-gating filters out.
3. **Breast mask segmentation.** The breast region is isolated using Otsu thresholding [21] on the uint8-converted image, followed by morphological closing and opening with an 11-pixel elliptical kernel, extraction of the largest connected component, and hole filling. This produces a binary mask separating the breast from the background.
4. **Contrast enhancement.** Contrast-Limited Adaptive Histogram Equalization (CLAHE) [22] is applied within the breast mask using a clip limit of 2.0 and a tile grid size of 8×8 , via OpenCV [23]. This locally enhances tissue contrast while preventing noise amplification.

Storage format. Preprocessed images and the extracted mass / background patches are saved as NumPy `.npy` arrays rather than the PNG files the reference paper uses. We checked that this choice does not materially affect downstream results: a controlled comparison on the 115-patch WS mass-classification task — feeding the Subspace k -NN ensemble with WS features extracted from `.npy` patches vs. an 8-bit PNG round-trip of the same patches (float $[0, 1] \rightarrow \text{uint8} \rightarrow 8\text{-bit grayscale PNG} \rightarrow \text{reload} \rightarrow \text{float } [0, 1]$) — produced *identical* held-out test accuracy on all five train/test split seeds tested ($\{1, 7, 34, 42, 99\}$), with cross-validation accuracy and AU-ROC differences well inside the across-seed noise floor (e.g. ~ 0.2 pp on CV-mean vs. a ~ 2 pp across-seed standard deviation, with the two non-zero deltas pointing in opposite directions on different metrics). At the pixel level the 8-bit PNG round-trip introduced a per-pixel MAE of $\sim 0.08\%$ — a half-step of 8-bit quantisation, as expected — which the Wavelet Scattering transform’s spatial-averaging step then absorbs, leaving a feature-vector relative MAE of only $\sim 0.02\%$. We therefore retain `.npy` as a practical implementation choice (no quantisation step, exact preservation of mask dtypes, and the native

format expected by Kymatio, PyTorch and scikit-learn), not as an accuracy improvement. The full numerical evidence is in `data/outputs/png_vs_npy_investigation/`.

Differences from the reference paper. The paper’s preprocessing includes a density-aware histogram adjustment technique (their reference [73], an earlier work by the same authors) rather than CLAHE. Their method adaptively adjusts contrast based on the clinical density category of the breast, which may provide better separation of tissue types. Our use of standard CLAHE, while effective as a general-purpose enhancement, does not incorporate density-specific information and likely constitutes one source of performance difference between our replication and the reported results. Additionally, the paper converts images to PNG format and uses seeded region growing for pectoral muscle segmentation, whereas we operate directly on DICOM arrays and use ground-truth polygon masks from an independent public source. Table 3.1 summarises the key differences.

Table 3.1 Preprocessing pipeline comparison between the reference paper and this replication.

Step	Paper [5]	This replication
Normalisation	DICOM $\rightarrow$ PNG (implicit min-max windowing); rescale to $[0, 1]$	1st/99th-percentile clip; rescale to $[0, 1]$
Image format	DICOM $\rightarrow$ PNG conversion	Operate directly on DICOM arrays
Background removal	Morphological operations; background set to NaN (MATLAB convention)	Otsu threshold + morphology + largest component + hole filling; background set to 0
Pectoral muscle removal	Seeded region growing (MLO views)	Ground-truth LabelMe JSON polygon masks [19] (202 MLO views)
Contrast enhancement	Density-aware histogram adjustment [73] (adapts to ACR category)	CLAHE (clip limit = 2.0, tile grid 8×8); density-agnostic
Patch tissue labelling	Otsu threshold on remaining breast region	Otsu threshold + ACR-guided class filtering

3.3 Patch Extraction and Tissue Labelling

Background tissue patches for the fatty versus fibroglandular classification task are extracted in *Step 1.3 — Background Patch Sampling*. The procedure operates on the 343 images that have XML annotations:

1. **Mass exclusion.** All ROIs labelled “Mass” in the XML annotations are parsed into polygon masks and dilated by 24 pixels to create an exclusion zone, ensuring that sampled background patches do not overlap with lesion tissue.
2. **Tissue segmentation.** Within the remaining allowed region (breast mask minus pectoral muscle minus dilated mass), an Otsu threshold splits tissue into a darker fatty region and a brighter fibroglandular region. A small gap of 0.03 around the threshold is enforced to reduce ambiguous patches at the boundary.
3. **ACR-guided sampling.** Unlike the paper, which does not describe density-aware patch filtering, we use the ACR breast density score to restrict which tissue classes each image may contribute. Images with ACR 1 (almost entirely fatty) contribute only fatty patches; ACR 4 (extremely dense) contributes only fibroglandular patches; and ACR 2–3 contribute both. This prevents mislabelling that can occur when Otsu thresholding forces a binary split on a breast that is clinically almost entirely one tissue type.
4. **Random patch sampling.** Up to 3 patches per allowed class per image are sampled at random positions and sizes (proportional to the breast bounding box, ranging from 14% to 30% of the shorter dimension, capped between 160 and 512 pixels). Each candidate patch must satisfy: full containment within the allowed mask, at least 65% of pixels belonging to the target tissue class, and a minimum pixel standard deviation of 0.035 to reject flat or near-empty patches. Overlapping patches within the same image are forbidden. All random choices (patch position and size) are drawn from a single seeded generator (`np.random.default_rng(34)`), so the sampled patch set is fully deterministic and identical on every run.
5. **Resizing.** Accepted patches are resized to 224×224 pixels using area-based interpolation and saved as float32 NumPy arrays.

This procedure yields 434 background patches in total: 251 fatty and 183 fibroglandular. The reference paper reports 397 patches (210 fatty, 187 fibroglandular); the difference arises from our ACR-guided sampling strategy and the absence of a hard-coded stopping criterion. Table 3.2 compares the patch counts side-by-side.

3.4 Wavelet Scattering Transform

The wavelet scattering transform provides a mathematically principled, non-learned feature representation that is locally invariant to translations and stable to small deformations [3, 4].

Table 3.2 Number of extracted patches. The paper uses $n = 112$ source images from INbreast; our pipeline processes all 343 images with XML annotations.

Task	Class	Paper [5]	This replication
Background tissue	Fatty	210	251
	Fibroglandular	187	183
Mass	Malignant	76	75
	Benign	36	41
Total		509	550

Mass patches are used in the benign-vs-malignant task reported in Section 4. The BI-RADS mapping used is 2–3 $\rightarrow$ benign and 4–6 $\rightarrow$ malignant; the paper does not state its exact cut-off, which accounts for the 41/36 difference in benign counts.

For an input signal x , the zeroth-order scattering coefficient is the local average $S_0x = x \star \phi_J$, where ϕ_J is a low-pass filter at scale 2^J . The first-order coefficients are obtained by convolving with a bank of oriented wavelets ψ_{λ_1} (parametrised by scale and orientation), applying the modulus non-linearity, and averaging:

$$S_1x = |x \star \psi_{\lambda_1}| \star \phi_J.$$

The second-order coefficients capture higher-frequency information lost by the first-order modulus, computed as:

$$S_2x = ||x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}| \star \phi_J.$$

Concatenating S_0 , S_1 , and S_2 produces the scattering feature vector.

Implementation. We use the Kymatio library [24] with the following configuration, chosen to align with the reference paper:

- Maximum scale: $J = 6$, corresponding to an invariance scale of $2^J = 64$ pixels.
- Number of orientations: $L = 5$ (Morlet wavelets [12]).
- Maximum scattering order: 2.
- Input shape: 224×224 .

Kymatio returns scattering coefficient *maps* of shape (C, H_s, W_s) ; following the paper’s approach, we spatially average each coefficient map to obtain one scalar per scattering path, yielding a compact 406-dimensional feature vector per patch.

Justification of the order and invariance-scale choices. Two of these settings are directly supported by the foundational scattering analysis of Bruna and Mallat [4]. First, the choice of maximum order 2 is not arbitrary: they show that scattering energy decays exponentially with path length, with over 99% of the energy carried by paths of length $m \leq 3$, and that second-order coefficients recover precisely the higher-moment texture information that the first-order (SIFT-like) coefficients discard — enough, for example, to discriminate textures with identical Fourier power spectra. Orders beyond 2 add negligible energy at substantial computational cost, so order 2 is the standard and well-motivated choice. Second, the large invariance scale ($2^J = 64$ on 224×224 patches) is appropriate because the fatty-versus-fibroglandular task is essentially a *texture* discrimination problem: Bruna and Mallat show that for stationary-texture classification the optimal invariance scale approaches the image (here, patch) width, since this minimises the variance of the windowed-scattering estimator. Treating each background patch as a texture sample therefore favours a large J , and the near-complete spatial pooling that results (Kymatio returns a 3×3 spatial grid per path at this setting) is consistent with that view rather than a loss of useful information.

Feature count discrepancy. The paper reports 391 features using MATLAB’s `waveletScattering` toolbox, which uses 6 orientations. The count follows the formula $1 + JL + \binom{J}{2}L^2 = 1 + 6 \times 6 + 15 \times 36 = 577$; however, the MATLAB toolbox applies $L = 6$ orientations and $J_{\max} = 5$ wavelet scales (since one scale is reserved for the low-pass filter), giving $1 + 5 \times 6 + \binom{5}{2} \times 6^2 = 391$. In Kymatio with $L = 5$ orientations and $J = 6$ scales (all 6 used for wavelets), the count is $1 + 6 \times 5 + \binom{6}{2} \times 5^2 = 406$. This minor difference is an unavoidable consequence of different library conventions. Table 3.3 compares the two configurations.

Table 3.3 Wavelet scattering configuration: reference paper (MATLAB) vs. this replication (Python / Kymatio).

Parameter	Paper [5]	This replication
Library / toolbox	MATLAB <code>waveletScattering</code>	Kymatio [24]
Wavelet type	Morlet	Morlet
Invariance scale	$64 (2^J \text{ pixels})$	$64 (2^J \text{ pixels})$
J (max decomposition)	6	6
Effective wavelet scales	5 (1 reserved for low-pass)	6 (all used)
L (orientations)	6	5
Max scattering order	2	2
Input size	224×224	224×224
Spatial pooling	Spatial averaging	Spatial averaging
Output features	391	406
Feature standardisation	None (raw coefficients)	<code>StandardScaler</code>

3.5 Classifier: Subspace k-NN Ensemble

Following the paper’s specification, we use a *Subspace k-Nearest Neighbours Ensemble* [15, 14] with the following hyperparameters:

- Number of ensemble learners: 80.
- Subspace dimensionality: 190 features (randomly selected per learner from the full 406-dimensional feature space).
- Number of neighbours: $k = 1$.
- Random seed: 34.

Each learner independently projects the data onto a random 190-dimensional subspace, fits a 1-nearest-neighbour classifier, and produces a class probability estimate. The final prediction is the soft vote (average probability) across all 80 learners. This ensemble approach improves robustness by averaging over many diverse views of the high-dimensional feature space. Beyond matching the reference paper, this choice is also well-suited to our small sample sizes: Bruna and Mallat observe that low-capacity / generative classifiers tend to outperform more flexible discriminative classifiers (such as kernel SVMs) when training samples are scarce, the discriminative classifiers only overtaking them as the training set grows [4]. A simple distance-based ensemble is therefore a reasonable estimator for our tasks, and especially for the small ($n \approx 115$) mass set. Table 3.4 lists the classifier hyperparameters.

Table 3.4 Subspace k-NN ensemble classifier settings. The paper tested several values and selected the optimal ones shown; we adopt the same optimal values.

Parameter	Values tested [5]	Paper optimal	Ours
Ensemble method	Subspace, AdaBoost, RUSBoost	Subspace	Subspace
Learner type	Nearest neighbour, Discriminant	Nearest neighbour	Nearest neighbour
Number of learners (k)	40, 60, 80, 100	80	80
Subspace dimension	170, 190, 210	190	190
Number of neighbours	—	1	1

The feature matrix is standardised using `StandardScaler` from `scikit-learn` [16] (zero mean, unit variance) before classification, as k -NN is distance-based and sensitive to feature scaling. Note that the paper does not mention applying standardisation to the WS features before classification; this is a difference in our pipeline that may affect results.

Train–test split. The 434 patches are split 80/20 into training and test sets, stratified by class, with a fixed random seed of 34 for reproducibility. The split is performed at the *image*

level (on unique `file_id` values) rather than at the patch level, so that all patches extracted from the same mammogram remain together in either the training or the test partition. This yields 344 training patches (199 fatty, 145 fibroglandular) and 90 test patches (52 fatty, 38 fibroglandular). Cross-validation likewise uses scikit-learn’s `StratifiedGroupKFold` [16] with `file_id` as the group label. The rationale for adopting image-level splitting, and the issue that prompted the change, is described in Section 4 (Stage 2).

3.6 Mass ROI Extraction

For the benign versus malignant mass classification task, mass ROIs are obtained from the INbreast XML annotations (the OsiriX plist files described in Section 3.1). Each annotation supplies a bounding box for the lesion. The paper refers generically to “the mass detection algorithm” but does not specify which detector; we therefore use the dataset’s expert annotations directly, which is the standard usage of INbreast. The procedure, implemented in *Step 1.2 — Mass ROI Extraction*, is:

1. Parse the XML bounding box for each “Mass” ROI, yielding (y_0, y_1, x_0, x_1) coordinates.
2. Crop the corresponding region from the already-preprocessed full image at `data/outputs/preprocessed` so the mass patch inherits the same percentile-clip normalisation, breast mask, pectoral removal, and CLAHE enhancement applied to the background patches.
3. Resize the crop to 224×224 pixels using area-based interpolation, clip values to $[0, 1]$, and save as a float32 NumPy array.

The label is assigned from the BI-RADS assessment: BI-RADS 2–3 $\rightarrow$ benign (class 0), and BI-RADS 4a/4b/4c/5/6 $\rightarrow$ malignant (class 1). This is the standard clinical convention. The paper does not state its exact mapping, which is a minor source of ambiguity. The procedure yields 116 mass patches (75 malignant + 41 benign); the paper reports 112 (76 + 36). A fuller reconciliation of this small discrepancy — which we attribute mainly to our retention of all `Name="Mass"` ROIs (including the few images with multiple mass ROIs) and to the paper’s apparent inclusion of some asymmetry/distortion findings as “mass tissue” rather than to the BI-RADS threshold — is given in Section 6.6.

3.7 CNN Branch: ResNet18 Fine-Tuning and Feature Extraction

The paper’s CNN branch (Model 1 in their Table 6) uses ResNet18 [18] pre-trained on ImageNet, with the final fully connected layer replaced to output two classes. The paper deliberately selects the comparatively shallow ResNet18 over deeper residual networks (ResNet50/101): with only a few hundred training patches per task, a shallower network is less prone to overfitting and to the information loss that accumulates in very deep architectures on small datasets [25]. We replicate this in PyTorch via `torchvision.models.resnet18(weights=IMAGENET1K` with the FC layer redefined as a $512 \rightarrow 2$ linear projection. All layers are fine-tuned (no freezing), matching MATLAB’s default `trainNetwork` behaviour. To meet ResNet’s 3-channel input requirement, each single-channel patch is replicated three times and normalised with the standard ImageNet mean and standard deviation. No data augmentation is applied, in line with the paper.

The hyperparameters follow Table 4 of the paper exactly:

- Optimiser: Adam, learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$.
- L_2 weight decay: 1×10^{-4} .
- Loss: cross-entropy.
- Number of epochs: 60.
- Mini-batch size: 80.

After fine-tuning, 512-dimensional feature vectors are extracted from the average-pooling layer of the trained network via a forward hook — i.e. from the penultimate layer, before the final FC. These features are then standardised with `StandardScaler` and classified by the same Subspace k-NN ensemble described in Section 3.5, matching the paper’s pipeline for Model 1.

Validation set for overfitting tracking. In addition to the 80/20 train–test split, we carve a $\sim 20\%$ validation set from the training images (image-level) and track its loss and accuracy at each epoch. This is an additional diagnostic not described in the paper, included after the supervisor flagged a concern about CNN overfitting; the diagnostic results are presented in Section 4 (Stage 4 and Stage 5b). The validation patches are excluded from gradient updates but re-included for the subsequent k-NN feature extraction so that the classifier sees the same training set as the paper.

3.8 Region-aware WS-map CNN head (extension)

This extension, developed for the mass task, keeps the spatial scattering maps and conditions them on lesion morphology rather than collapsing them to a scalar vector. It has four ingredients.

Region masks from polygon ROIs. For each mass, the OsiriX XML Name="Mass" polygon is rasterised to a binary lesion mask on the preprocessed image. From this single outline, four lesion-relative regions are derived by morphological operations: a *full* region (the lesion with a small surrounding margin), an *inner* region (the eroded core), a *boundary* region (a dilated-minus-eroded annulus straddling the margin), and a *context* region (a dilated surround with the lesion removed). The erosion/dilation structuring elements are sized as a fraction of the lesion bounding box, so the decomposition scales with lesion size. Each region mask is applied to the patch, giving four region images per lesion.

$J = 3$ scattering maps. Each region image is passed through the same Kymatio scattering transform but at $J = 3$ rather than the paper's $J = 6$. This retains a 28×28 spatial grid of 91 coefficient channels per region, instead of collapsing to the near-scalar 3×3 grid that $J = 6$ produces. Stacking the four regions gives a 4×91 -channel, 28×28 scattering-map tensor per lesion that preserves *where* coefficient energy sits within each lesion sub-region.

WS-map CNN head. A small convolutional network is trained directly on these scattering-map tensors: two 3×3 convolutional blocks (Conv-BatchNorm-ReLU, 32 channels), global average pooling, dropout (0.5), and a linear classifier, optimised with Adam ($\eta = 10^{-3}$, weight decay 3×10^{-4}) for 60 epochs at batch size 24. The CNN learns spatial relationships across the region maps that a hand-pooled scalar summary discards; this is the main methodological extension of the project.

Comparators. Two controlled comparators accompany the WS-map CNN. (i) A *classical* region-aware baseline pools each region map to per-channel statistics and classifies the concatenated vector with logistic regression or the Subspace k-NN ensemble, isolating the value of the region decomposition from the value of the learned head. (ii) An *improved CNN* comparator extracts 512-d features from a *frozen* ImageNet ResNet18 on a square lesion crop and classifies them with logistic regression — a stable, fine-tuning-free CNN baseline. Finally, *late* (decision-level) fusion averages the predicted malignancy probabilities of the region-aware WS→CNN and the improved CNN, in contrast to the paper's feature-level concatenation.

3.9 Final evaluation protocol and reporting

The development narrative in Section 4 reports each stage on a single fixed image-level split (seed 34) with group-aware 10-fold cross-validation, reflecting how the experiments were originally run. For the *final* comparison of the headline methods this is not enough on its own: the mass held-out set contains only ~ 25 patches from ~ 22 images, so a single additional misclassification shifts test accuracy by ~ 4 pp, and a single split can swing by 10 pp or more between seeds. The final evaluation protocol therefore follows four principles.

1. **Repeated, image-level splits.** Each headline method is evaluated over five repeated stratified train/test splits (seeds $\{1, 7, 34, 42, 99\}$), each grouped by `file_id` so that no patches or ROIs from the same mammogram fall in both the training and test partitions. Reporting the spread across seeds, rather than one fixed split, exposes the split sensitivity that a single number conceals.
2. **Leakage-free feature extraction.** Where a method uses a trained feature extractor (the CNN branch), the network is *retrained inside each split* on that split’s training images only, so the test features come from a model that never saw them.
3. **Training-only scaling.** The `StandardScaler` is fitted on the training fold only and then applied to the test fold, so that no test statistics influence training.
4. **AUROC-led reporting under class imbalance.** The mass task is imbalanced (74 malignant vs. 41 benign, a majority-class baseline of $\sim 68\%$), so raw accuracy is misleading — a trivial “always malignant” rule already scores $\sim 68\%$. We therefore lead with the area under the ROC curve (AU-ROC), which is threshold-independent and unaffected by class prevalence, and report accuracy, precision, recall, and F1 alongside it. All metrics are quoted as mean $\pm$ standard deviation across the five seeds.

Five seeds is a deliberate trade-off: enough to reveal the across-seed variability while keeping every headline method on a common, reproducible set of splits. Because all seeds resample the same ~ 115 mass patches, the reported standard deviation should be read as an indicator of split sensitivity on a small dataset, not as a tight confidence interval. Cross-validation is not used as a headline metric: the repeated held-out splits already give an out-of-sample estimate, and cross-validation computed on *learned* CNN features is optimistic because the feature extractor has already seen the inner-validation images. The consolidated comparison under this protocol is produced by the canonical *FINAL Step 2.1–2.5* notebooks.

Reproducibility. All randomness is pinned per run: the Python, NumPy and PyTorch seeds are set together, and wherever a CNN is trained, cuDNN is placed in deterministic mode (`torch.backends.cudnn.deterministic=True, benchmark=False`) so that

repeated runs of the same seed agree. The canonical diagnostic seed is 34; the headline numbers additionally average the five image-level seeds $\{1, 7, 34, 42, 99\}$. For the learned WS-map CNN head, which is trained on only ~ 90 patches, we report a five-seed mean $\pm$ standard deviation and corroborate it with a more stable group-aware cross-validation rather than relying on any single split (Chapter 6).

Chapter 4

Paper-Style Replication Results

This section presents the classification results in the chronological order in which the experiments were conducted. Each stage records what was tried, the result obtained, what the result suggested, and the decision that followed. Tables 4.1 and 4.2 at the end of the section consolidate the comparison against the paper’s reported benchmarks. The stage-by-stage numbers below are reported on the single fixed seed-34 image-level split (the development setting in which the experiments were run); the consolidated tables report the final *five-seed* mean $\pm$ standard deviation protocol (Section 3.9), which is the headline. Where a single-split figure is unusually high or low because of the small mass test set, the five-seed value is given alongside it. This development narrative refers to the original exploratory notebooks by their chronological names (*Step 2.1 — WS*, *Step 2.2 — CNN*, *Step 2.3 — Fusion*, *Step 2.8 — Consolidated Mass Evaluation*); their methods and results were subsequently consolidated into the canonical *FINAL Step 2.1–2.5* notebooks — tissue replication in *FINAL Step 2.1*, mass replication in *FINAL Step 2.2*, and the extensions in *FINAL Step 2.3–2.5* — with the exploratory originals retained under *Archive/*. All headline tables in this report are drawn from the FINAL notebooks.

4.1 Stage 1 — Wavelet Scattering Baseline (Fatty vs Fibroglular)

What was tried. The first experiment implements the paper’s Model 2 (WS-only) on the background tissue task: 434 patches passed through the Kymatio scattering transform configured as in Section 3.4, spatially averaged to a 406-dimensional vector per patch, standardised, and classified by the Subspace k-NN ensemble. Cross-validation uses 10 folds.

Result. Seed-34 group-aware 10-fold cross-validation accuracy was $81.2\% \pm 7.3\%$ (the value carried into the master table); held-out test accuracy on the seed-34 split was 84.4% (Precision 0.800, Recall 0.842, F1 0.821, AU-ROC 0.926).

What this suggests. The replication is qualitatively consistent with the paper (which reports $\sim 91\%$) but trails by roughly 7 pp. The most likely sources are (i) our use of standard CLAHE in place of the paper’s density-aware histogram adjustment, and (ii) minor library differences in the scattering transform (Kymatio’s 406 features versus MATLAB’s 391; Table 3.3).

Decision. The WS-only baseline reproduces the paper’s qualitative pattern but trails the reported number. Two follow-ups were planned: (i) implement the CNN branch (Section 4.2 onwards), and (ii) later return to the WS branch to explore an alternative scattering representation suggested by the project supervisor. The latter is reported as the scattering covariance extension (Section 5.2) in the Extension Experiments chapter, although in calendar time it was carried out after the CNN, mass, and fusion stages.

4.2 Stage 2 — First CNN Attempt and Diagnosis of Data Leakage

What was tried. A first ResNet18 fine-tune, exactly per Section 3.7, was run on the fatty versus fibroglandular task. The train–test split used at this stage was the obvious patch-level interpretation of the paper’s wording “the data is divided into a randomized ratio of eight-to-two” — i.e. `train_test_split` on the flat patch index, stratified by class.

Result. Every reported metric came out at 100%: training accuracy, 10-fold cross-validation accuracy, and held-out test accuracy. This was implausible.

Investigation. An audit of which `file_ids` appeared in each partition revealed that, of approximately 100 source images contributing background patches, 59 had patches in both the training and the test set. Of the 87 test patches, 86 (98.9%) shared a source mammogram with at least one training patch. Because patches from the same mammogram are highly correlated — same patient, same view, same density characteristics, often spatially adjacent regions of the same tissue — this constituted near-complete information leakage from training to test.

What this suggests. The paper’s wording is ambiguous between patch-level and image-level splitting. Their reported CNN-only test accuracy of 77.4% alongside a 99% cross-validation accuracy (Table 6 of [5]) is in fact more consistent with image-level splitting: a

patch-level split would tend to produce CV and test numbers that match each other rather than show a large gap. We had inadvertently chosen the wrong interpretation.

Decision. Switch to image-level splitting and group-aware cross-validation for *both* branches and *both* tasks. Re-run the WS Stage 1 baseline with the corrected splits (the numbers reported in Section 4.1 above are post-correction).

4.3 Stage 3 — Image-Level Splitting and Group-Aware Cross-Validation

What changed. Both `train_test_split` and cross-validation were modified to operate on the unique `file_id` (source mammogram) rather than the patch index. The cross-validator was changed from `StratifiedKFold` to `StratifiedGroupKFold` with `groups=file_id`, which preserves stratification by class while guaranteeing that all patches from the same image stay in the same fold. The 80/20 ratio and the random seed (34) were preserved.

Why. To prevent the within-fold and train–test leakage diagnosed in Stage 2, and to give an honest estimate of how the model generalises to genuinely unseen patients.

Effect. The post-correction WS Stage 1 numbers were already reported above. The CNN replication, repeated with the same fix, is presented in Stage 4.

4.4 Stage 4 — CNN-Only Tissue Classification (Corrected Splits)

What was tried. The CNN branch described in Section 3.7 was retrained with image-level splitting and a validation set carved from the training images. The fine-tuned ResNet18 was evaluated in two ways: (a) directly via the softmax head, and (b) by extracting 512-d features from the average-pool layer and classifying them with the same Subspace k-NN ensemble used for WS (this is the paper’s Model 1 protocol).

Result — CNN direct (softmax). Train accuracy 100.0%, validation accuracy 100.0%, test accuracy 97.8%, test AU-ROC 0.994. The overfit gap (train – test) was 2.2%.

Result — CNN → Subspace k-NN (paper protocol). 10-fold group-aware CV $100.0\% \pm 0.0\%$, test accuracy 97.8%, AU-ROC 0.977, Precision 0.974, Recall 0.974, F1 0.974. Training time 330.7 s.

What this suggests. The CNN branch substantially *outperforms* the paper’s reported 77.4% test accuracy on the same task. Two observations are worth highlighting. First, the overfit gap is modest ($\sim 2\%$), much less severe than the 21.6% CV-vs-test gap the paper

reports (their narrative is that this gap is what motivates the CNN + WS fusion). Second, the validation *loss* is volatile mid-training, with brief drops in validation accuracy near epochs 25–30 before recovery (visible in the per-epoch training/validation curves logged by the notebook). This pattern indicates mild but real overfitting that is masked here by a relatively easy decision boundary.

The most likely reason our test accuracy substantially exceeds the paper’s is a cleaner pre-processing pipeline: percentile-clipped normalisation, ground-truth pectoral muscle masks, CLAHE, and ACR-guided patch selection together provide the network with less ambiguous training data than the paper’s MATLAB pipeline. This is consistent with the WS branch trailing by ~ 7 pp on the same task, since WS reads texture directly from the same preprocessing whereas the CNN can in principle exploit any additional signal made cleaner by it.

Decision. Tissue classification is now replicated for both branches. Proceed to the mass classification task (the paper’s second experiment, Table 7).

4.5 Stage 5 — Extending the Pipeline to Mass Classification

What was tried. The benign versus malignant mass classification task (Section 3.6) was added to both notebooks as new final sections, leaving the tissue sections untouched. The 116 mass patches were built from the XML bounding boxes (75 malignant, 41 benign; the paper has 76 + 36). The image-level 80/20 split yields 91 training patches across 85 images and 25 test patches across 22 images. Both the WS-only and CNN-only protocols from the tissue task were re-applied unchanged.

Stage 5a — WS-Only Mass Classification

Result. 10-fold group-aware cross-validation accuracy was $74.7\% \pm 8.8\%$; held-out test accuracy was 52.0% (Precision 0.667, Recall 0.588, F1 0.625, AU-ROC 0.599). This is a single seed-34 split and is highly variable at $n_{\text{test}} \approx 24$: the leakage-free five-seed mean is markedly higher and more stable — AU-ROC 0.726 ± 0.030 , test accuracy 0.701 ± 0.075 (Table 4.2). The 52.0% single-split figure should therefore not be read as the model’s representative performance.

What this suggests. The cross-validation number is close to the paper’s reported 69% and is in fact slightly better. The held-out test accuracy, however, is far below the paper’s 83.3%. With only 25 test patches and a 32%/68% benign/malignant split, a single additional misclassification moves test accuracy by 4 pp, so the test estimate is high-variance. The discrepancy is also qualitatively consistent with the paper’s own observation that mass

tissue is a less linearly separable target for WS than background tissue: they describe mass classification as a setting where “less usable features of WS could be utilized.”

Stage 5b — CNN-Only Mass Classification

What was tried. A *fresh* ResNet18 was fine-tuned on the mass patches (not warm-started from the tissue model), with the same hyperparameters and the same train/val/test image-level partitioning. The validation set is 17 images (18 patches). Training took 25.6s for 60 epochs.

Result — CNN direct. Train accuracy 100.0%, validation accuracy 77.8%, test accuracy 84.0%, test AU-ROC 0.934. Overfit gap (train – test) was 16.0%.

Result — CNN → Subspace k-NN (paper protocol). 10-fold group-aware CV $95.6\% \pm 5.4\%$, test accuracy 80.0%, AU-ROC 0.779, Precision 0.875, Recall 0.824, F1 0.849.

What this suggests. All six headline metrics in Table 7 (paper’s CNN-only column) reproduce to within 1–3 pp of the paper’s reported values (paper: Test 0.833, CV 1.000, AUC 0.80, P 0.889, R 0.800, F1 0.842). The per-epoch training and validation accuracy curves logged by the notebook exhibit the same overfitting signature the paper describes: training accuracy saturates by epoch 10 while validation accuracy oscillates in the 60–90% range. This is the diagnostic the supervisor asked for, and it confirms the paper’s narrative: a pure CNN trained on this small a sample size overfits and would benefit from handcrafted-feature fusion.

4.6 Stage 6 — CNN + WS Feature Fusion (Both Tasks)

What was tried. The paper’s headline model (Model 5 in their Table 6; the “CNN + WS” column of their Table 7) fuses the CNN and WS branches by *concatenating* the two feature vectors per patch and feeding the combined vector to the same Subspace k-NN ensemble used by the single-feature models. No feature selection, no dimensionality reduction, and (per supervisor instruction) no GLCM branch were applied. The implementation, in *Step 2.3 — Fusion*, joins the saved WS features (406-d per patch, from Step 2.1) and CNN avgpool features (512-d per patch, from Step 2.2) on the `patch_npy` key to guarantee per-patch alignment. The labels and the train/test split columns are asserted to match exactly between the two source CSVs before concatenation, and the assertions pass. CNN features for the mass task were not persisted by Step 2.2 and are re-extracted in Step 2.3 by reloading the saved `resnet18_benign_malignant.pth` weights. The fused vector has dimension $512 + 406 = 918$. `StandardScaler` is fitted on the training fold(s) only.

Result — Tissue (Table 6 Model 5 equivalent). 10-fold group-aware CV $100.0\% \pm 0.0\%$; test accuracy 98.9%; AU-ROC 0.987; Precision 1.000; Recall 0.974; F1 0.987. Training time 4.9s (the CNN backbone is already trained; this is just the k-NN ensemble fit on the fused vectors).

Result — Mass (Table 7 “CNN + WS” equivalent). 10-fold group-aware CV $95.6\% \pm 5.4\%$; test accuracy 88.0%; AU-ROC 0.824; Precision 0.889, Recall 0.941, F1 0.914. Training time 10.8s. These are the seed-34 single split; under the leakage-free five-seed protocol the fusion settles to AU-ROC 0.816 ± 0.057 , test accuracy 0.792 ± 0.058 (Table 4.2), and the CV figure is optimistic because the CNN features were extracted before the CV split.

What this suggests. On the tissue task, fusion gives a small but real improvement over the CNN-only baseline (98.9% vs 97.8% test; one fewer misclassified patch out of 90) and exceeds the paper’s reported 93.5% by a similar margin to the CNN-only result, consistent with the cleaner preprocessing pipeline used throughout this replication. On the mass task, fusion gives a substantially larger boost over CNN-only on the seed-34 split (88.0% vs 80.0% test, +8 pp; the recall climbs from 0.824 to 0.941 and the F1 from 0.849 to 0.914). This is the regime the paper’s narrative emphasises: handcrafted scattering features add genuine new information when the CNN is overfitting on a small sample. The F1 and recall scores fall within ~ 1 pp of the paper’s Table 7 values for the same configuration. The seed-34 test AU-ROC (0.824 vs the paper’s 1.00) is the one number that lags, attributable to the very small test set ($n_{\text{test}} = 25$) and the resulting coarse probability ranking; the seed-99 five-seed mean (0.816 ± 0.057 , Table 4.2) is the stable estimate, and only a larger held-out set would tighten it further.

Decision. The paper’s main contribution has been reproduced for both tasks. No further single-method replication work is planned within the scope of this project; remaining improvements are write-up and robustness work (Section 6).

4.7 Cross-Method Summary

Tables 4.1 and 4.2 summarise all replicated results against the paper’s reported benchmarks. Numbers in the “Ours” rows are taken directly from the saved CSV outputs of the FINAL notebooks (data/outputs/final_tissue_replication/ and data/outputs/final_mass_replication/).

4.8 Why an extension to the WS branch is needed

Two facts from the replication motivate the extension experiments of Chapter 5. First, the *tissue* task is essentially solved by the paper-style pipeline: the CNN (AU-ROC 0.979) and

Table 4.1 Fatty vs. fibroglandular classification: paper benchmarks versus our replication. “Ours” rows use the final five-seed protocol (image-level seeds $\{1, 7, 34, 42, 99\}$): AU-ROC and test accuracy are mean $\pm$ standard deviation, and CV is the seed-34 group-aware 10-fold accuracy kept only as a paper-comparison metric. Paper values are from Table 6 of [5]. Source: `final_tissue_replication/master_table_tissue.csv`.

Method	AU-ROC	Test Acc.	CV (seed 34)	F1
<i>Paper [5] — MATLAB (single split)</i>				
WS-only	0.94	0.914	0.911	0.910
CNN-only	0.90	0.774	0.990	0.808
CNN + WS	0.96	0.935	0.993	0.934
<i>This replication — Python / Kymatio / PyTorch (five-seed)</i>				
WS-only (averaged, Mallat)	0.895 ± 0.070	0.839 ± 0.071	0.812 ± 0.073	0.807
CNN-only (ResNet18 $\rightarrow$ k-NN)	0.979 ± 0.010	0.979 ± 0.010	1.000 (opt.)	0.976
CNN + WS fusion (concat)	0.988 ± 0.007	0.964 ± 0.030	1.000 (opt.)	0.957
<i>Extension (Chapter 5)</i>				
Adapted scattering cov. ($J=6, L=3$)	0.984 ± 0.010	0.959 ± 0.013	0.962 ± 0.023	0.951

CV is 10-fold (group-aware in our replication, standard stratified in the paper). “(opt.)” marks an optimistic CV: the learned CNN features were extracted before the CV split, so this CV is not an honest out-of-sample estimate (Section 3.9).

the CNN + WS fusion (0.988) are near-saturated and the averaged WS branch is already respectable (0.895), so there is little headroom left to chase by reworking the tissue pipeline. Second, the *mass* task is both harder and more fragile: the held-out set is tiny (24 patches), the across-seed spread is wide, and — most importantly — the paper-style *averaged* wavelet scattering is weak (AU-ROC 0.726, only just below the CNN-only 0.762 and well below fusion’s 0.816).

The averaging is the prime suspect. Collapsing each scattering map to a single scalar discards *where* lesion structure sits, which is exactly the core-versus-margin-versus-context information that distinguishes benign from malignant masses (Chapter 2); on a near-homogeneous background-tissue patch nothing is lost, but on a structured lesion a great deal is. The natural question is therefore whether the WS branch can be improved by (a) a richer *scalar* descriptor, or (b) *preserving* the spatial maps and the lesion’s internal structure. The next chapter pursues both: it first screens a broad set of supporting, non-region-aware extensions (normalisation, statistical and pooled WS, scattering covariance, CNN crops, and fusion variants), finds their gains useful but limited, and then commits, for the mass task, to preserving the scattering spatial map and learning over it with a small CNN — with a lesion-relative region-aware

Table 4.2 Benign vs. malignant mass classification: paper benchmarks versus our replication. “Ours” rows use the final five-seed protocol (mean $\pm$ standard deviation; AU-ROC-led under class imbalance), with seed-34 group-aware CV retained only as a paper-comparison metric. Paper values are from Table 7 of [5]. Source: `final_mass_replication/master_table_mass.csv`.

Method	AU-ROC	Test Acc.	CV (seed 34)	F1
<i>Paper [5] (single split)</i>				
WS-only	0.80	0.833	0.690	0.842
CNN-only	0.80	0.833	1.000	0.842
CNN + WS	1.00	0.917	0.980	0.918
<i>This replication — five-seed, post data-quality filter (115 patches)</i>				
Majority baseline	0.500 ± 0.000	0.668 ± 0.031	0.633 ± 0.051	0.801
WS-only (averaged, Mallat)	0.726 ± 0.030	0.701 ± 0.075	0.711 ± 0.113	0.781
CNN-only (ResNet18 $\rightarrow$ k-NN)	0.762 ± 0.117	0.766 ± 0.067	0.989 (opt.)	0.829
CNN + WS fusion (concat)	0.816 ± 0.057	0.792 ± 0.058	0.978 (opt.)	0.842
<i>Extension (Chapter 5)</i>				
Adapted scattering cov. ($J=6, L=3$)	0.734 ± 0.062	0.682 ± 0.115	—	0.732

Held-out set is the post-filter 24-patch split (Section 6.7). “(opt.)” marks optimistic CV on learned features. Scattering covariance gives *no* AU-ROC gain over averaged WS on mass — it is a tissue-side extension (Chapter 5); the mass gains come from the region-aware methods. The earlier single-seed figures (e.g. WS-only test 0.52, fusion test 0.88) were high-variance single splits and are superseded by these five-seed means.

decomposition as its interpretable instantiation — as the project’s main methodological extension.

Chapter 5

Extension Experiments

This chapter collects the extension work carried out after the paper-style replication of Chapter 4, organised around the two tasks. After a broad *screening* of supporting, non-region-aware methods (Section 5.1, from *FINAL Step 2.3*), it follows two threads. On the *tissue* side, the *scattering covariance* descriptor (Section 5.2) is the main WS-branch improvement — a richer *scalar* representation that suffices for texture. On the *mass* side (Section 5.3, from *FINAL Step 2.4*), the consistent lesson — previewed in Section 4.8 — is that what rescues the weak averaged-WS branch is *preserving the scattering spatial map and learning over it with a small CNN*, rather than swapping scalar features or classifiers; the lesion-relative *region-aware* decomposition is the interpretable instantiation of that idea. Supporting CNN-crop and fusion experiments (Section 5.4) and a layer of negative, unstable, or superseded controls preserved in *FINAL Step 2.5* (summarised in Section 5.5) document the main extension search. The final section (Section 5.6) reports a late preprocessing-sensitivity investigation: it does not replace the main replication, but it shows that removing CLAHE is a useful mass-side sensitivity for WS and region-aware WS-map methods. Unless explicitly labelled otherwise, all numbers use the locked five-seed image-level protocol of Section 3.9.

5.1 Screening of supporting extensions

Before committing to a deep extension, a broad set of non-region-aware methods was screened under the locked five-seed protocol (*FINAL Step 2.3*). Table 5.1 summarises them, with mass AU-ROC as the headline metric (the imbalanced task; averaged-WS baseline 0.726, majority baseline 0.500). The recurring pattern is that changing the *scalar* feature or the classifier moves mass performance only modestly and within across-seed noise, whereas the one method that *preserves the spatial map* and learns over it — the full-patch WS→CNN head (0.909) — jumps clear of the strict paper-style WS branch, motivating the region-aware

deep dive (Section 5.3). On the *tissue* side, several structured summaries already saturate the task, and scattering covariance remains the main adopted WS-specific extension (Section 5.2).

Project-wide master comparison

For a more complete view, Tables 5.2–5.3 consolidate the main paper-replication, supporting-extension, region-aware, and appendix-level methods into task-specific master comparisons. They are intentionally AU-ROC-led, matching the final protocol and the discussion of class imbalance in Section 3.9. The companion plots in Figures 5.1–5.2 show the same comparison visually with across-seed error bars where available. The full generated source table is saved as `data/outputs/report_figures/all_methods_master.csv`; the printed tables below keep the report readable by focusing on the report-level and most informative negative-control rows.

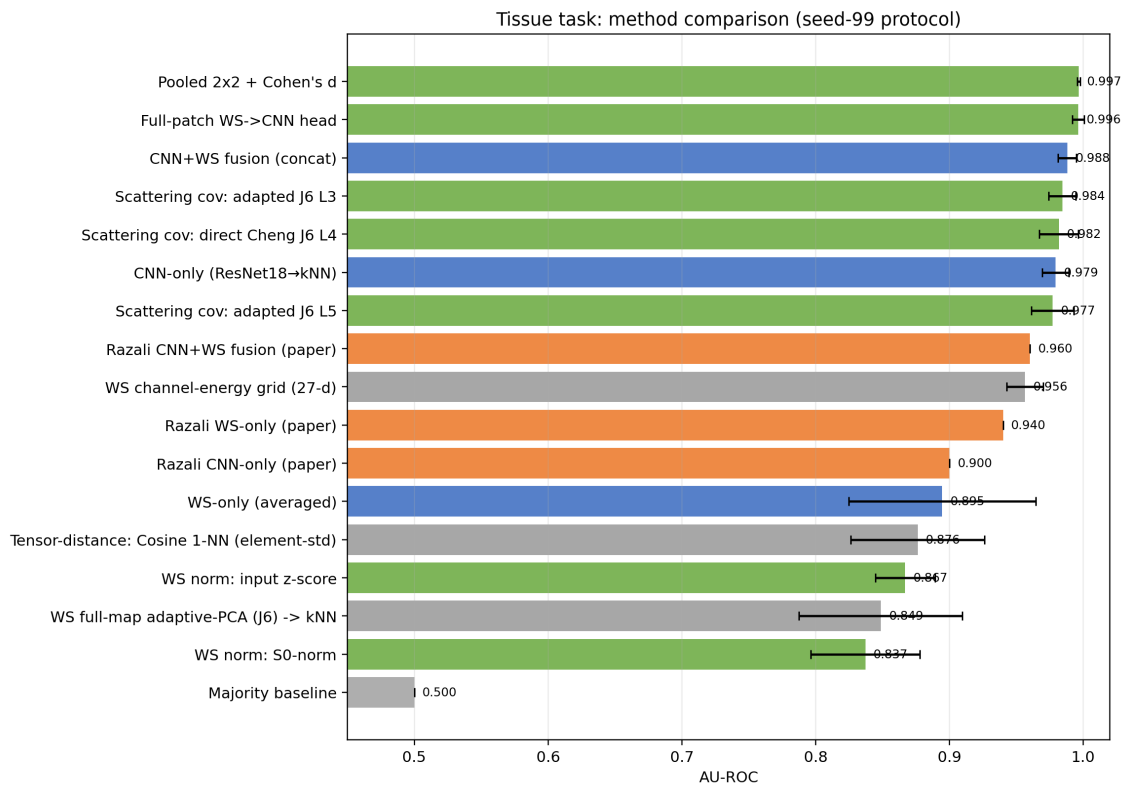


Fig. 5.1 Tissue task master comparison as a horizontal AU-ROC bar chart. Error bars show across-seed standard deviation for project methods; Razali paper rows are single reported values. The main visual lesson is that tissue is close to saturated for several structured representations, with scattering covariance and CNN/fusion methods all near the top.

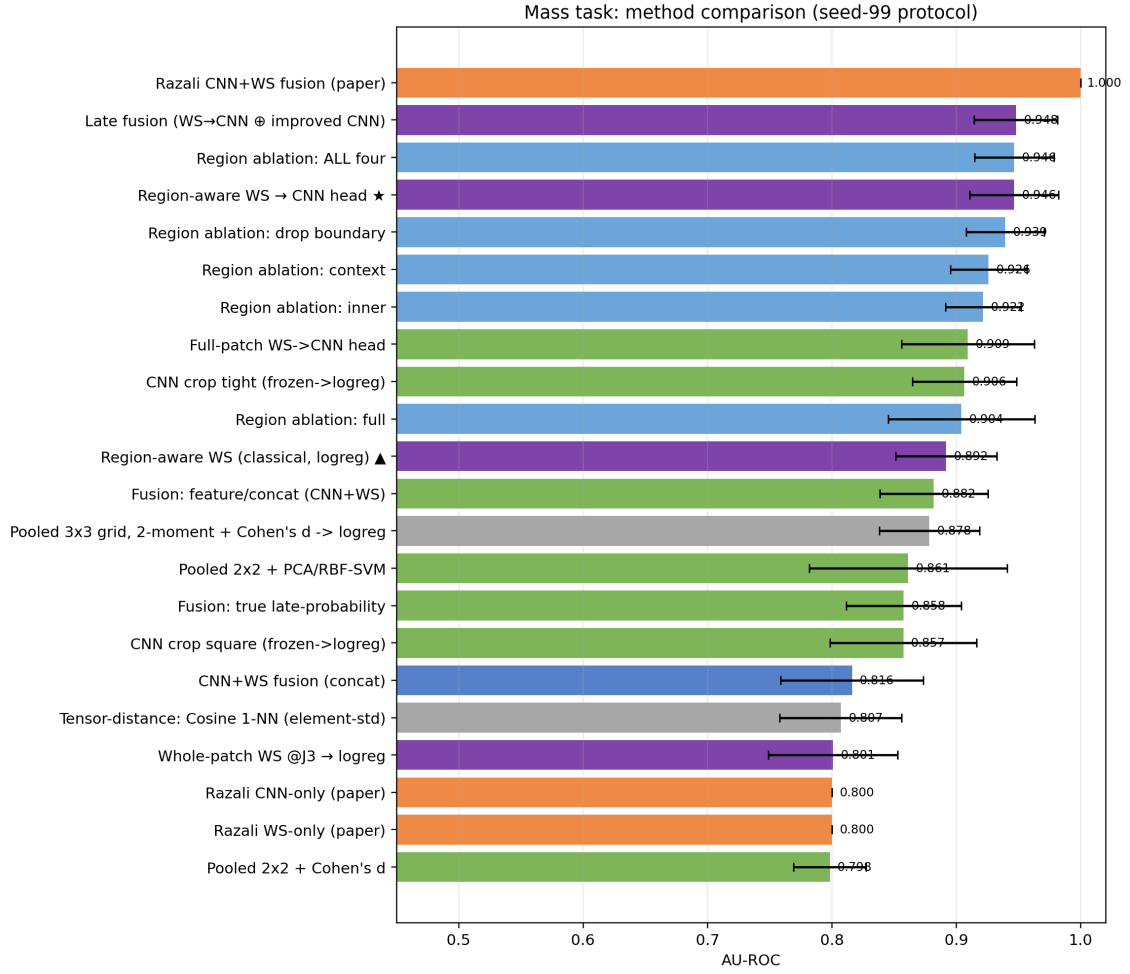


Fig. 5.2 Mass task master comparison as a horizontal AU-ROC bar chart. Error bars show across-seed standard deviation for project methods; Razali paper rows are single reported values. Unlike the tissue task, scalar WS variants cluster near the averaged-WS baseline, while spatial-map-preserving WS→CNN and its late fusion variants form the strongest group.

5.2 Scattering Covariance as the Tissue-Side Extension (Supervisor-Suggested)

Motivation

The WS-only baseline (Section 4.1) trails the paper's reported result by roughly 7 pp on the tissue task and, as later stages will show, gives a relatively weak mass-classification result. With strict paper-replication established as a reference point, the project supervisor recommended exploring the scattering covariance methodology of Cheng & Ménard [8] as a richer texture representation. The supervisor explicitly warned against the alternative of

simply flattening the full scattering tensor and applying PCA — a non-canonical operation that would conflict with the mathematical construction of the scattering transform. We therefore investigated scattering covariance as an extension that:

- adds higher-order cross-correlation statistics on top of the standard S_0, S_1, S_2 coefficients;
- is mathematically principled (defined in terms of spatial averages of products of scattering fields, not flattened spatial maps);
- is directly applicable to texture / field analysis, which matches the mammogram patch setting.

This direction is consistent with the foundational scattering analysis: Bruna and Mallat [4] already show that it is the *second-order, cross-path* structure of the scattering representation — not the first-order coefficients alone — that carries the discriminative higher-moment texture information (for instance, the information needed to separate textures with identical power spectra). Scattering covariance can be seen as making this cross-path structure explicit through dedicated correlation statistics, so the extension builds on, rather than departs from, the theory underpinning the Stage 1 baseline.

This work was approached as an extension of, not a replacement for, the paper-aligned averaged-WS baseline of Section 4.1. That baseline remains the faithful Razali replication; the present section explores whether the WS branch can be improved beyond it.

Scattering covariance — what it is and why we used it

The standard Mallat-style scattering transform used in Stage 1 retains the zeroth-, first-, and second-order spatial averages $S_n = \langle I_n(\vec{x}) \rangle$ of the cascaded wavelet-modulus fields (Section 3.4). Cheng & Ménard’s scattering covariance adds further *scalar* summary statistics computed from products of these scattering fields — principally P_{00} (power per wavelet band), C_{01} (cross-correlations between the input and its modulus fields), C_{11} (cross-correlations between pairs of modulus fields), and P_{11} (autocorrelations of modulus fields). Each of these is defined as a spatial average of a product of scattering fields, so the output is a tensor of scalars indexed by scale and orientation tuples, never a stack of spatial maps. The supervisor’s no-flattening constraint is therefore satisfied by construction: there is no tensor to flatten.

For an isotropic stochastic texture (which mammogram patches roughly are), Cheng’s paper recommends the *rotationally-averaged* vector `for_synthesis_iso`: {mean, std, $\log P_{00\text{iso}}$, $\log S_{1\text{iso}}$, $C_{01\text{iso}}$,

with first-order orientation averaged out and second-order orientation differences retained. This is the scalar feature vector we use throughout this section.

Adapted vs. direct usage. Two configurations were evaluated. *Adapted* runs use the Cheng `scattering_cov` API with parameters matched to the project’s existing WS configuration ($J = 6, L = 5$ initially, later tuned — see the ablation below). *Direct* runs use the same API with $L = 4$, the default value in the Cheng package’s example script. Both use the same Subspace k-NN ensemble (80 learners, dim 190, 1-NN) on top, the same image-level 80/20 train/test split, and the same group-aware 10-fold cross-validation as Stage 1, so the comparison is on equal footing.

Initial four-way comparison (tissue task)

The first scattering covariance experiment compared four methods on the fatty-vs-fibroglandular task: the original WS (Stage 1), the adapted scattering covariance at $J = 6, L = 5$, the direct Cheng usage at $J = 6, L = 4$, and the paper’s reported WS-only number (Razali Table 6 Model 2, reproduced from publication).

What the comparison shows. All scattering covariance variants substantially exceed the original averaged WS on tissue (AU-ROC $0.895 \rightarrow 0.977\text{--}0.984$; test accuracy $0.839 \rightarrow 0.923\text{--}0.959$). The variants differ mainly in the orientation parameter L , so rather than simply adopting whichever scored highest on one metric we tuned J and L systematically (below), ranking by the more stable seed-34 group-aware CV.

Parameter ablation — tuning J and L for scattering covariance

The scattering transform has two principal hyperparameters: J (maximum dyadic scale; 2^J is the invariance scale in pixels) and L (number of wavelet orientations). The Cheng paper does not prescribe specific values; Razali used the MATLAB toolbox defaults ($J = 6, L = 6$) without ablation; the Cheng example script uses $L = 4$ on 512-pixel inputs. Given that the project’s goal had shifted from strict replication to obtaining the best defensible scattering-based result, we performed a small grid search over J and L , ranking candidates by 10-fold group-aware CV mean (the most stable estimator on this sample size) with CV standard deviation as a tiebreaker. The search range was deliberately small and physically motivated; both axes were swept on the training partition only (no test-set peeking).

L sweep at fixed $J = 6$. The L -sweep produces a clear picture. Cross-validation accuracy is highest at $L = 3$ (0.962 ± 0.023), with $L = 1$ and $L = 2$ within ~ 0.6 pp on CV but at higher variance once $L \geq 5$. From $L = 4$ onwards CV declines, and the held-out AU-ROC

peaks at $L = 3$ (0.984). Feature count grows from 90 at $L = 1$ to 3,765 at $L = 8$, and the high-dimensional settings coincide with higher CV variance, consistent with overfitting in the small-sample regime. $L = 3$ is the most stable, parsimonious choice that achieves the top CV mean.

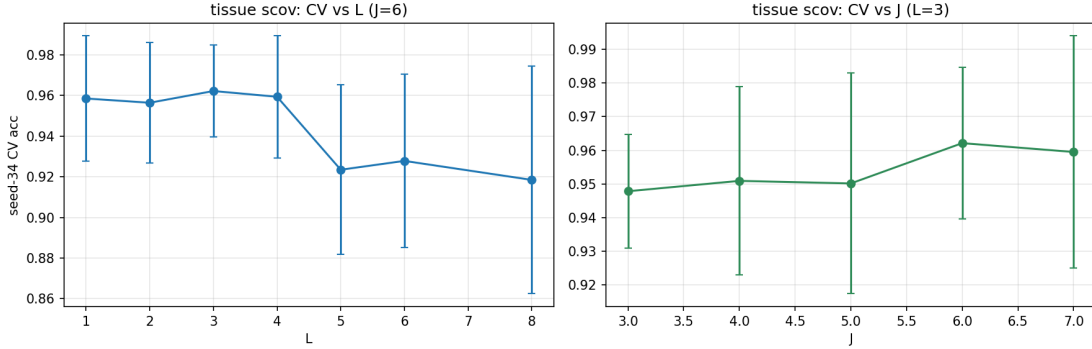


Fig. 5.3 Tissue scattering-covariance ablation (*FINAL Step 2.3*): seed-34 group-aware CV accuracy versus L at fixed $J = 6$ (left) and versus J at fixed $L = 3$ (right). CV peaks at $L = 3$ and $J = 6$; larger L raises dimensionality and CV variance without improving accuracy.

J sweep at fixed $L = 3$. The J -sweep is flatter than the L -sweep. CV clusters between 0.948 and 0.962, with $J = 6$ at the top; held-out AU-ROC rises gently with J (to 0.984 at $J = 6$ and 0.985 at $J = 7$) while CV variance grows. The choice $J = 6$ gives the best CV with the lowest variance among the leading settings.

Final adapted configuration. The ablation selects $J = 6, L = 3$ as the project’s adapted scattering covariance setting. It wins on CV mean (0.962), gives the lowest CV standard deviation among the top candidates (0.023), a strong five-seed held-out AU-ROC (0.984) and test accuracy (0.959), and the smallest feature count of the leading configurations (580, $\sim 60\%$ smaller than the original $L = 5$ setting). The choice is data-driven via CV-based selection on the training partition; it deviates from Razali’s MATLAB-default $L = 6$ but is well justified by the small-sample regime, where higher-dimensional scattering representations give less stable nearest-neighbour distances.

Mass task: scattering covariance gives no AU-ROC gain

The chosen adapted configuration was also applied to the benign-vs-malignant mass task under the same five-seed protocol (Table 5.7). Here the tissue success does *not* transfer.

The mass picture is the opposite of the tissue picture. The best scattering-covariance setting on mass ($J = 6, L = 3$, AU-ROC 0.734 ± 0.062) is statistically indistinguishable from

the averaged-WS baseline (0.726 ± 0.030), and the other settings are lower. The richer scalar covariance therefore does *not* solve the mass task: it reshapes the precision/recall trade-off but lowers F1 relative to the baseline (0.732 vs. 0.781) and adds no meaningful discriminative power by AU-ROC.

A small mass-side L -control at $J = 6$ confirms this is not a tuning artefact (Table 5.8): no orientation count rescues mass scattering covariance — the best, $L = 3$, merely returns to the averaged-WS baseline. This is an L -only control at $J = 6$, *not* a full mass $J \times L$ grid; a full grid would over-invest in a representation that is, on this task, a dead end. The mass gains instead come from preserving spatial/lesion structure (Section 5.3), not from richer scalar statistics.

The earlier development notebooks reported a single-seed mass result that appeared to show a covariance “jump” (test 0.708 against an unusually low averaged-WS single split of 0.542); under the seed-99 five-seed protocol that apparent jump disappears, as the averaged WS recovers to 0.701 test / 0.726 AU-ROC. We therefore treat scattering covariance as a *tissue-side* extension only, and any comparison to Razali’s reported mass number remains not apples-to-apples (different library, preprocessing, patch counts, and a ~ 24 -patch held-out set).

Conclusion: scattering covariance is the tissue-side WS extension

The supervisor-suggested scattering covariance extension delivers a clear, defensible improvement over averaged WS *on the tissue task*, and is adopted as the project’s main WS-branch tissue representation:

Adapted scattering covariance, $J = 6$, $L = 3$, for_synthesis_iso, with StandardScaler + Subspace k-NN ensemble.

This configuration:

- gives the highest CV mean (0.962) and lowest CV variance (0.023) of any WS-branch method tested for the tissue task, with five-seed held-out AU-ROC 0.984;
- exceeds the averaged WS baseline on tissue by +0.089 AU-ROC (0.895 $\rightarrow$ 0.984) and +0.120 test accuracy (0.839 $\rightarrow$ 0.959);
- gives *no meaningful* AU-ROC gain on mass (0.734 vs. the 0.726 averaged-WS baseline), so it is adopted on the tissue side only;
- respects the supervisor’s no-flattening constraint by construction (the output is a scalar coefficient vector, not a flattened spatial tensor);

- is parsimonious (580 features, versus 1,518 at $L = 5$ and the averaged WS's 406);
- is selected by CV-based hyperparameter tuning on the training partition only, with no test-set peeking.

The Direct Cheng implementation ($J = 6, L = 4$) is retained as an independent sensitivity check, confirming the tissue result is not specific to the $L = 3$ choice; the Mallat-scattering implementation remains the faithful paper-aligned baseline. Both feed the cross-method summary (Section 4.7).

Decision. Scattering covariance is adopted as the improved WS representation *for tissue*; the averaged WS continues to serve as the paper-replication reference. Because it does not help mass, the mass task is taken up separately by the region-aware extension (Section 5.3). Reproducing CNN+WS fusion on top of scattering-covariance features is identified as natural further work but is out of the current scope.

5.3 Region-aware wavelet scattering (mass)

The project's main mass-side extension is to *preserve the scattering spatial map and learn over it with a small CNN (FINAL Step 2.4)*. Its premise (Section 4.8) is that averaged WS fails on mass because it discards *where* lesion structure lies; rather than enrich a scalar descriptor — as scattering covariance does for tissue — this approach keeps the scattering *maps* at a lower invariance scale and lets a small CNN learn over them. The lesion-relative *region-aware* decomposition described below is the interpretable, morphology-conditioned instantiation of that idea; as the ablation will show, the measured accuracy comes mainly from preserving the map and learning over it, with the region split adding interpretability rather than a proven numerical gain.

Construction

The machinery is described in Section 3.8: each lesion's XML polygon is rasterised to a mask, from which four lesion-relative regions (*full*, *inner* core, *boundary*/margin annulus, and surrounding *context*) are derived by erosion/dilation; each region is scattered at $J = 3$ (retaining 91 channels on a 28×28 grid); the four region maps are stacked (4×91 channels) and fed either to pooled classical classifiers or to a small WS-map CNN head. Keeping $J = 3$ rather than the paper's $J = 6$ is the key choice: it preserves a usable spatial grid instead of collapsing each map to a near-scalar.

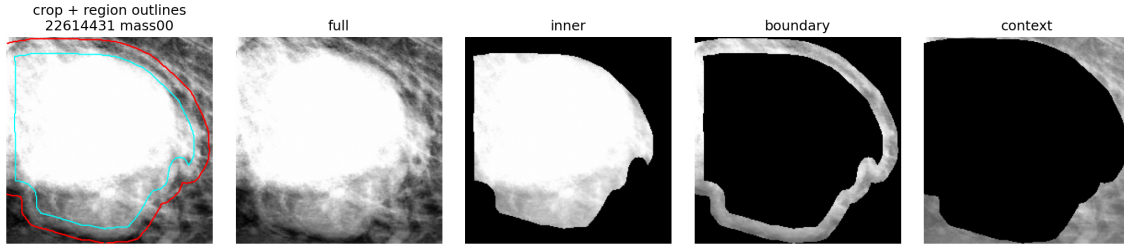


Fig. 5.4 Region-aware decomposition of a single mass lesion (malignant; file_id 22614431). From the lesion mask, four lesion-relative region images are formed *before* wavelet scattering: the *full* whole-lesion crop, the *inner* eroded core, the *boundary* margin band (dilation minus erosion), and the surrounding *context*. The leftmost panel shows the crop with the eroded-core outline (cyan) and the dilated lesion-plus-margin outline (red). Each region is scattered at $J = 3$ and the four region maps are stacked, so the descriptor is conditioned on lesion morphology rather than averaged over the whole patch.

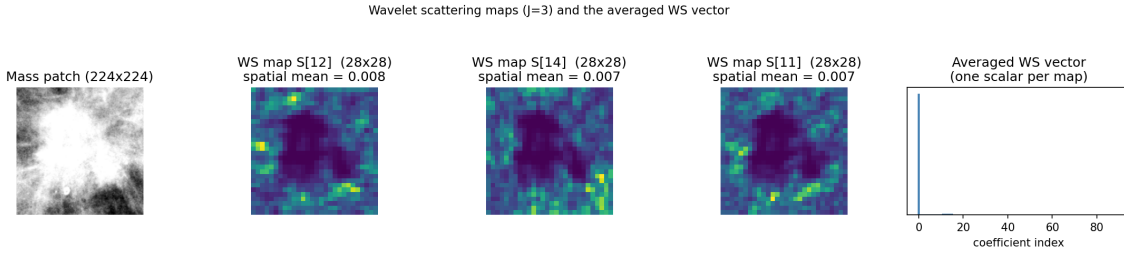


Fig. 5.5 Why paper-style averaging discards spatial information. For an example mass patch (left), the wavelet scattering transform at $J = 3$ produces 91 coefficient maps of size 28×28 (three shown, centre), each encoding *where* that scale/orientation responds within the lesion. The paper-style averaged WS keeps only the spatial *mean* of each map — one scalar per coefficient (right) — collapsing all 28×28 structure to a point. The region-aware extension instead retains these maps and conditions them on lesion regions.

Step-by-step results

Table 5.9 walks one change at a time from the paper-style baseline to the region-aware models, all under the five-seed image-level protocol; Figure 5.6 plots the same progression.

The progression is monotone and interpretable. Moving from $J = 6$ averaged WS to $J = 3$ maps with a linear head already raises AU-ROC from 0.726 to 0.801 — spatial detail helps even on the whole patch. Conditioning on lesion regions raises the classical (logistic-regression) model further to 0.892, and letting a small CNN learn over the region maps reaches 0.946 ± 0.036 . Under the updated five-seed protocol, this all-four-region WS-map CNN is the strongest *standalone* mass-side model in the project; only the later decision-level late fusion is numerically higher, and that is an ensemble of two trained branches. A small CNN over the *whole-patch* $J = 3$ map — with no region decomposition at all — reaches

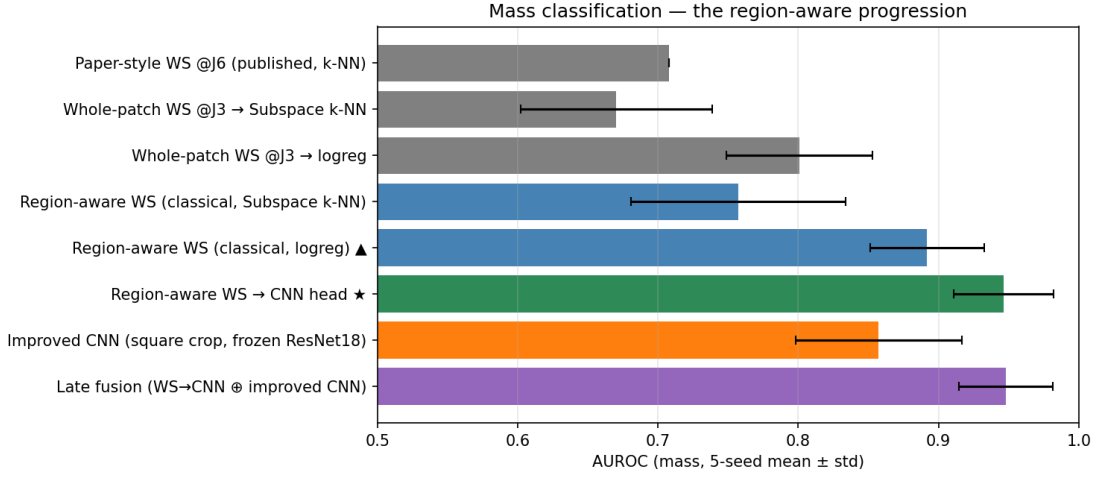


Fig. 5.6 Mass task, region-aware extension story (five-seed AU-ROC). Each step changes one thing: lowering the invariance scale to $J = 3$ and switching to a linear head lifts whole-patch WS from 0.726 to 0.801; decomposing the lesion into regions lifts the classical model to 0.892; a learned WS-map CNN head reaches 0.946; and a decision-level late fusion with a frozen-ResNet comparator reaches 0.948.

0.909 ± 0.053 (Table 5.11), so the decisive step remains preserving the map and learning over it; the lesion-region decomposition now gives an additional numerical lift while also making the descriptor morphology-aware.

A region ablation (*FINAL Step 2.4*, Table 5.10) tests what the four-region decomposition actually contributes. Every configuration that retains a $J = 3$ map and learns over it sits far above the averaged-WS baseline (0.726), but the strongest rows are now the complete all-four-region tensor (0.946), the full+inner pair (0.940), and the leave-one-out variant that drops the boundary band (0.939). The strong configurations still overlap within across-seed spread, so the region ordering should not be over-read. The robust conclusion is that *preserving the spatial map and learning over it* is the main lever; using all four lesion-relative regions is a principled and currently strongest complete representation because it encodes the lesion core, margin, whole lesion, and surrounding context.

Stability of the learned head.

Because the WS-map CNN is a *learned* model trained on only ~ 90 patches, its five-seed held-out AU-ROC still carries split sensitivity (± 0.036). A group-aware 10-fold cross-validation diagnostic gives a tighter estimate, AU-ROC 0.938 ± 0.007 , confirming that the held-out mean is not a lucky split. We report both, and treat the held-out five-seed mean as the headline with the group-CV as the stability check (Section 6).

Master comparison.

Table 5.11 places the region-aware models against the paper-style replication and the strongest supporting extensions on the mass task.

Interpretation. Late fusion is the numerically strongest configuration in the table (0.948), but it is a decision-level *combination* of two models, and on this small mass set its margin over the region-aware WS→CNN model alone (0.946, group-CV 0.938) is negligible relative to across-seed noise. We therefore do *not* present late fusion as the project’s core contribution. The methodological contribution is *preserving the wavelet-scattering spatial map and learning over it with a small CNN*: this is what turns the weak averaged-WS branch (0.726 AU-ROC) into a strong mass descriptor — the whole-patch $J = 3$ -map CNN reaches 0.909, and the lesion-region decomposition lifts this to 0.946 while adding morphological interpretability. The classical pooled variants, the improved-CNN comparator, and late fusion support that story rather than replace it.

5.4 Supporting mass extensions: CNN crops and fusion

Alongside the WS-representation work, two CNN-side levers were screened on mass (*FINAL Step 2.3*); both support the region-aware story rather than supplant it.

Crop / context sweep with a frozen ResNet18. The fine-tuned ResNet18 of the replication overfits the ~ 90 mass patches and swings widely across seeds, so a *frozen* ImageNet ResNet18 was used as a fixed 512-d feature extractor with a logistic-regression head, varying only the crop:

Crop	AU-ROC	Test Acc.
Tight bounding box	0.906 ± 0.042	0.815 ± 0.039
Square crop	0.857 ± 0.059	0.808 ± 0.054
Expanded ($1.5\times$) context	0.820 ± 0.042	0.766 ± 0.078

The frozen extractor is far more stable than the fine-tuned model, and logistic regression beats RBF-SVM and Subspace k-NN on these features. The tight box scores the highest AU-ROC, but tight and square overlap within the across-seed spread; the region-aware comparison (Section 5.3) uses the *square* crop as its “improved CNN” comparator (0.857 ± 0.059) because it is the steadier controlled crop used throughout that notebook. Either way this is a stable comparator, not a mass solution.

Whole-patch fusion vs. region-aware fusion. On the whole patch, two fusions of the frozen CNN with averaged WS were compared: feature/early fusion (concatenate the

512-d CNN and 406-d WS vectors, then one classifier) at AU-ROC 0.882 ± 0.043 , and true late-probability fusion (average the two branches' predicted probabilities) at 0.858 ± 0.046 . Feature fusion is the stronger of the two, but neither reaches the region-aware spatial-map branch. This is a *different* construction from the region-aware late fusion of Section 5.3, which combines the much stronger region-aware WS→CNN (0.946) with the improved CNN and reaches 0.948 ± 0.033 . Late fusion helps there precisely because one of its branches already preserves lesion structure: fusion amplifies a good spatial branch but cannot manufacture one from averaged features.

5.5 Wavelet scattering alternatives (appendix-level)

A further set of WS variants was tried and is preserved, with full provenance, in the appendix notebook *FINAL Step 2.5*; none was adopted, so they are summarised here only briefly. Two *flattening* approaches — reshaping the scattering map into one long vector, and flatten-then-PCA — were run explicitly as negative controls, because the supervisor had advised against flattening the scattering tensor; their results did not justify adoption, confirming the guidance. *Tensor-distance* classifiers (1-NN under Frobenius/cosine distances on the standardised scattering tensor, and a channel-subspace vote) did not outperform the simple spatial-mean baseline and were markedly slower. Finally, *J-sweep* and grid-granularity experiments showed that the productive direction is *lower-J* spatial maps fed to a learned head — exactly the route taken by the full-patch WS→CNN (Section 5.1) and the region-aware WS→CNN (Section 5.3) — rather than richer flattening or distance metrics. These appendix methods document the breadth of the search and the negative results that justify the narrower set of methods reported above.

5.6 Preprocessing sensitivity: masked/no-CLAHE inputs

Motivation

After the paper-style replication and the region-aware extensions were complete, a final sensitivity question remained: how much of the WS behaviour was caused by the *amount of preprocessing* applied before patch extraction? The supervisor suggested using preprocessing primarily to identify coordinates and relevant regions, then extracting the classification patches from less-altered images. This was motivated by the fact that wavelet scattering is texture- and structure-sensitive: operations such as CLAHE can sharpen local contrast and

help visual inspection, but they may also alter the coefficient energy that scattering uses for classification.

This was treated as an extension and sensitivity analysis, not as a replacement for the paper-style replication. The main replication remains the faithful Razali-aligned pipeline; the aim here was to test whether a lighter input representation explains some of the remaining reproduction gap and whether it changes the mass-side region-aware extension.

Initial pixel-source screen

The first test reused the existing preprocessing-derived patch coordinates and labels, but changed the pixel source from which the patch was extracted. Four sources were compared in *FINAL Step 2.6*: the current fully preprocessed images; original DICOM pixels with simple min–max scaling (`raw_minmax`); original DICOM pixels with only 1st/99th percentile scaling (`raw_percentile`); and `masked_no_clahe`. The last variant means: 1st/99th percentile normalisation, breast/background masking, and pectoral-region removal, but *no* CLAHE contrast enhancement.

The useful effect was concentrated in the *mass WS-only* setting (Table 5.12). Tissue WS did not improve: the current full-preprocessing row remained closest to the Razali tissue WS reference (AU-ROC 0.895 vs. the paper’s 0.940), while raw or lighter variants fell to 0.881 or 0.812. CNN-only and CNN/fusion sensitivity checks were also inconclusive: some settings moved AU-ROC closer to the paper and others moved test accuracy closer, but no CNN variant matched both. This is why the remainder of the preprocessing investigation focuses on mass WS and region-aware WS-map methods.

By AU-ROC alone, raw min–max is closest to Razali (0.797 vs. 0.800). However, `masked_no_clahe` is the best balanced limited-preprocessing variant: it is still essentially tied to the paper’s AU-ROC (0.805 vs. 0.800) and gives the highest test accuracy among the tested alternatives. This suggests that CLAHE is the step most likely to be distorting mass WS texture information: keeping the anatomical masking while removing contrast enhancement preserves enough breast/pectoral cleanup without imposing an additional local contrast transform.

Region-aware WS-map CNN with masked/no-CLAHE inputs

Because the useful effect appeared in mass WS, the next test repeated the region-aware WS-map CNN pipeline with `masked_no_clahe` inputs. The setup was otherwise unchanged from Section 5.3: the same 115 mass samples, the same XML mass polygons, the same four lesion-relative regions (*full*, *inner*, *boundary*, *context*), the same $J = 3, L = 5$ spatial

scattering tensors, and the same small CNN head. Only the pixel source used to build the region images changed.

Table 5.13 shows that removing CLAHE improved the all-region model on every headline metric: AU-ROC rose from 0.946 to 0.971, test accuracy from 0.899 to 0.925, and F1 from 0.923 to 0.943. The standard deviations also narrowed, suggesting that the improvement is not only a single favourable split.

Full region ablation under masked/no-CLAHE

The full region ablation was then repeated under the same `masked_no_clahe` input style (Table 5.14). This is the same ablation grid used in Section 5.3: single regions, *full*+*X* pairs, leave-one-out triples, and all four regions.

The ablation strengthens, but also refines, the original region-aware conclusion. The broad conclusion still holds: the mass gain comes from preserving WS spatial maps and learning over them. Under masked/no-CLAHE, however, the strongest AU-ROC is obtained by *dropping the boundary* (0.986 ± 0.014), i.e. using full, inner, and context regions but excluding the margin band. This is consistent with the idea that the boundary region is the most sensitive to preprocessing artefacts: it is precisely where lesion edge, surrounding tissue, interpolation, and contrast enhancement interact. Removing CLAHE appears to make the spatial maps cleaner overall, but the boundary channel can still add noise for AU-ROC.

At the same time, the all-four-region model gives the best balance of test accuracy and F1 (0.925 and 0.943 respectively). Therefore the cautious reading is metric-dependent: *drop boundary* is the best AU-ROC configuration, while *all four regions* remains the most balanced and interpretable representation. This is why the main extension narrative does not need to be rewritten around a single ablation winner. The final insight is instead that preprocessing is not neutral for WS features: removing CLAHE improves the mass WS branch and the region-aware WS-map CNN, especially when the model is allowed to learn from preserved spatial scattering maps.

Table 5.1 Screening of supporting (non-region-aware) extension methods, from *FINAL Step 2.3* (`supporting_extensions_results.csv`). AU-ROC is the five-seed mean; “status” records whether each method was adopted, pursued further, kept as supporting evidence, or dropped. Averaged-WS baselines: mass 0.726, tissue 0.895.

Method	Task	AU-ROC	Status	Lesson
<i>WS representation / classifier variants</i>				
WS normalisation (S0 / z-score)	mass	0.72/0.78	dropped	no consistent gain over averaged WS
$J=3$ per-path / statistical WS	mass	0.66	dropped	extra scalars do not beat the mean
Pooled 2×2 + Cohen’s d	mass	0.798	supporting	structured pooling + selection helps modestly
Pooled 2×2 + PCA/RBF-SVM	mass	0.861	supporting	best hand-pooled mass result
Scattering ($J=6, L=3$) covariance	tissue	0.984	adopted	strongest adopted tissue-side WS extension
Scattering ($J=6, L=3$) covariance	mass	0.734	dropped	no meaningful AU-ROC gain on mass
Full-patch WS $\rightarrow$ CNN head	mass	0.909	pursued	preserving the spatial map is the biggest lever
<i>CNN variants (frozen ResNet18, mass)</i>				
Tight crop $\rightarrow$ logreg	mass	0.906	supporting	highest crop AU-ROC (within noise of square)
Square crop $\rightarrow$ logreg	mass	0.857	supporting	stable comparator carried into Section 5.3
Expanded crop $\rightarrow$ logreg	mass	0.820	dropped	over-large crop dilutes the lesion
<i>Fusion variants (whole-patch, mass)</i>				
Feature/early fusion (concat)	mass	0.882	supporting	improves on the strict k-NN fusion head
True late-probability fusion	mass	0.858	supporting	no robust whole-patch fusion gain
<i>Appendix / negative controls (FINAL Step 2.5)</i>				
Flatten / flatten+PCA; tensor-distance	both	—	dropped	did not beat simpler baselines; against the no-flattening guidance

Table 5.2 Tissue task — master method comparison across paper benchmarks, replication models, extensions, and selected negative controls. AU-ROC is the five-seed mean $\pm$ standard deviation unless the row is a Razali paper benchmark.

Method	Family	AU-ROC	Test	Status / lesson
Pooled 2×2 + Cohen's d	WS extension	0.997 ± 0.001	0.975	strong but not primary
Full-patch WS $\rightarrow$ CNN head	spatial-map screen	0.996 ± 0.004	0.968	tissue already saturated
CNN + WS fusion (concat)	replication	0.988 ± 0.007	0.964	paper-style fusion
Scattering covariance ($J=6, L=3$)	WS extension	0.984 ± 0.010	0.959	adopted tissue WS extension
Direct Cheng scattering covariance	WS extension	0.982 ± 0.015	0.943	sensitivity check
CNN-only (ResNet18 $\rightarrow$ kNN)	replication	0.979 ± 0.010	0.979	near-saturated
Scattering covariance ($J=6, L=5$)	WS extension	0.977 ± 0.016	0.923	higher-dimensional variant
Razali CNN + WS fusion	paper benchmark	0.960	0.935	published reference
WS channel-energy grid	appendix control	0.956 ± 0.014	0.961	strong tissue-only control
Razali WS-only	paper benchmark	0.940	0.914	published reference
Razali CNN-only	paper benchmark	0.900	0.774	published reference
WS-only (averaged)	replication	0.895 ± 0.070	0.839	paper-style WS baseline
Tensor-distance cosine 1-NN	appendix control	0.876 ± 0.050	0.829	heavier, no gain
WS input z-score / S_0 normalisation	normalisation controls	$0.867 / 0.837$	$0.792 / 0.740$	dropped
WS full-map adaptive-PCA	appendix control	0.849 ± 0.061	0.801	flatten/PCA underperforms
Majority baseline	baseline	0.500	0.570	trivial reference

Table 5.3 Mass task — master method comparison across paper benchmarks, replication models, supporting extensions, region-aware/spatial-map models, and selected negative controls. AU-ROC is the five-seed mean $\pm$ standard deviation unless the row is a Razali paper benchmark.

Method	Family	AU-ROC	Test	Status / lesson
Razali CNN + WS fusion	paper benchmark	1.000	0.917	single-split reference
Region-aware late fusion	spatial-map / fusion	0.948 ± 0.033	0.883	numerically strongest
Region-aware WS→CNN head	region-aware extension	0.946 ± 0.036	0.899	interpretable all-region model
Region ablation: ALL four	region ablation	0.946 ± 0.032	0.899	strongest complete region set
Region ablation: drop boundary	region ablation	0.939 ± 0.031	0.858	highest leave-one-out row
Region ablation: context only	region ablation	0.926 ± 0.031	0.824	strongest single region
Full-patch WS→CNN head	spatial-map screen	0.909 ± 0.053	0.809	key spatial-map lever
CNN tight crop (frozen→logreg)	CNN screen	0.906 ± 0.042	0.815	strong comparator
Region-aware WS (classical logreg)	region-aware extension	0.892 ± 0.041	0.806	regions without learned head
Whole-patch feature fusion (CNN+WS)	fusion screen	0.882 ± 0.043	0.790	strongest whole-patch fusion screen
Pooled 3×3 + Cohen's d logreg	appendix control	0.878 ± 0.040	0.809	promising but noisy
Pooled 2×2 + PCA/RBF-SVM	WS screen	0.861 ± 0.080	0.792	best hand-pooled mass WS
Whole-patch late-probability fusion	fusion screen	0.858 ± 0.046	0.760	no robust whole-patch fusion gain
CNN square crop (frozen→logreg)	CNN screen	0.857 ± 0.059	0.808	stable comparator
CNN expanded crop	CNN screen	0.820 ± 0.042	0.766	context dilution
CNN + WS fusion (paper-style concat)	replication	0.816 ± 0.057	0.792	best strict replication model
Tensor-distance cosine 1-NN	appendix control	0.807 ± 0.049	0.734	heavier, no gain
Razali WS-only / CNN-only	paper benchmark	0.800 / 0.800	0.833 / 0.833	published references
Pooled 2×2 + Cohen's d	WS screen	0.798 ± 0.029	0.722	modest structured pooling gain
WS full-map flatten ($J=6$)	appendix control	0.788 ± 0.072	0.724	no reliable gain
WS input z-score normalisation	normalisation control	0.781 ± 0.042	0.750	dropped
CNN-only (ResNet18→kNN)	replication	0.762 ± 0.117	0.766	unstable small-sample CNN
Scattering covariance ($J=6, L=3$)	WS extension	0.734 ± 0.062	0.682	no meaningful mass AU-ROC gain
WS-only (averaged)	replication	0.726 ± 0.030	0.701	weak mass WS baseline
WS S_0 normalisation	normalisation control	0.721 ± 0.051	0.750	dropped
WS full-map adaptive-PCA	appendix control	0.684 ± 0.042	0.733	underperforms
Majority baseline	baseline	0.500	0.668	accuracy misleading

Table 5.4 Tissue task — WS comparison under the five-seed protocol (AU-ROC and test accuracy as mean $\pm$ std; CV is seed-34 group-aware accuracy). “Ours” rows use identical patches, splits, scaler, and classifier. The Razali row is the published WS-only result and is not apples-to-apples (different library, preprocessing, patch counts, CV scheme); it is a reference, not a head-to-head competitor. Source: `supporting_extensions_results.csv`, `scov_JL_ablation_tissue.csv`.

Method	AU-ROC	Test	CV (seed 34)	F1
Original WS (Kymatio, 406-d)	0.895 ± 0.070	0.839 ± 0.071	0.812	0.807
Adapted scattering cov. ($J=6, L=5$, 1518-d)	0.977 ± 0.016	0.923 ± 0.031	0.923	0.908
Adapted scattering cov. ($J=6, L=3$, 580-d)	0.984 ± 0.010	0.959 ± 0.013	0.962	0.951
Direct Cheng ($J=6, L=4$, 993-d)	0.982 ± 0.015	0.943 ± 0.024	0.959	0.932
Razali (2023) WS-only (reported, 391-d)	0.94	0.914	0.911	0.910

Table 5.5 L sweep at fixed $J = 6$ for the adapted scattering covariance, tissue task, five-seed protocol. CV is the seed-34 group-aware mean; AU-ROC and test accuracy are five-seed means. Source: `scov_JL_ablation_tissue.csv`.

L	# feat.	CV	CV std	AU-ROC	Test
1	90	0.958	0.031	0.950	0.950
2	279	0.956	0.030	0.978	0.959
3	580	0.962	0.023	0.984	0.959
4	993	0.959	0.030	0.982	0.943
5	1518	0.923	0.042	0.977	0.923
6	2155	0.928	0.043	0.975	0.918
8	3765	0.918	0.056	0.964	0.921

Table 5.6 J sweep at fixed $L = 3$, tissue task, five-seed protocol. CV is the seed-34 group-aware mean; AU-ROC and test accuracy are five-seed means. Source: `scov_JL_ablation_tissue.csv`.

J	# feat.	CV	CV std	AU-ROC	Test
3	115	0.948	0.017	0.928	0.931
4	219	0.951	0.028	0.958	0.941
5	371	0.950	0.033	0.971	0.941
6	580	0.962	0.023	0.984	0.959
7	855	0.960	0.035	0.985	0.957

Table 5.7 Mass task — scattering covariance vs. averaged WS, five-seed protocol (AU-ROC and test accuracy mean $\pm$ std; 115 patches, 24-patch held-out split). Source: `supporting_extensions_results.csv`.

Method	AU-ROC	Test	F1
Averaged WS (Mallat, 406-d)	0.726 ± 0.030	0.701 ± 0.075	0.781
Adapted scattering cov. ($J=6, L=5$, 1518-d)	0.645 ± 0.072	0.622 ± 0.079	0.695
Adapted scattering cov. ($J=6, L=3$, 580-d)	0.734 ± 0.062	0.682 ± 0.115	0.732
Direct Cheng ($J=6, L=4$, 993-d)	0.654 ± 0.055	0.617 ± 0.083	0.688

Table 5.8 Mass-side L -control for scattering covariance at fixed $J = 6$ (five-seed AU-ROC; negative control). No L meaningfully beats the averaged-WS baseline of 0.726. Source: `scov_L_ablation_mass.csv`.

L	# feat.	AU-ROC	Test
1	90	0.544 ± 0.103	0.609
2	279	0.642 ± 0.051	0.649
3	580	0.734 ± 0.062	0.682
4	993	0.654 ± 0.055	0.617
5	1518	0.645 ± 0.072	0.622

Table 5.9 Mass task: region-aware extension story, five-seed image-level protocol (AU-ROC mean $\pm$ std). Source: *FINAL Step 2.4*, `recovered_region_aware_ws_cnn/mass_extensions_story_table.csv`.

Method	AU-ROC	Test Acc.
Paper-style WS @ $J=6$ (averaged, k-NN)	0.726 ± 0.030	0.701
Whole-patch WS @ $J=3$ maps $\rightarrow$ k-NN	0.670 ± 0.068	0.709
Whole-patch WS @ $J=3$ maps $\rightarrow$ logreg	0.801 ± 0.052	0.733
Region-aware WS (classical, k-NN)	0.757 ± 0.077	0.789
Region-aware WS (classical, logreg)	0.892 ± 0.041	0.806
Region-aware WS $\rightarrow$ CNN head ★	0.946 ± 0.036	0.899
Improved CNN (square crop, frozen ResNet18)	0.857 ± 0.059	0.808
Late fusion (region-aware WS $\rightarrow$ CNN $\oplus$ improved CNN)	0.948 ± 0.033	0.883

Table 5.10 Region ablation for the WS→CNN head (five-seed AU-ROC; selected rows). Every configuration sits well above the averaged-WS baseline (0.726), but differences among the stronger ones are within seed noise. Source: `region_ws_ablation/region_ablation_cnnhead_results.csv`.

Region configuration	AU-ROC
ALL four (full+inner+boundary+context)	0.946 ± 0.032
Full + inner	0.940 ± 0.013
Drop boundary (full+inner+context)	0.939 ± 0.031
Full + boundary	0.933 ± 0.035
Drop context (full+inner+boundary)	0.931 ± 0.018
Context only	0.926 ± 0.031
Inner only	0.922 ± 0.030
Full region only	0.904 ± 0.059

Table 5.11 Mass task — master comparison across families, five-seed image-level protocol (AU-ROC mean $\pm$ std), drawn from the FINAL Step 2.1–2.4 outputs. “Status” indicates the method’s role in the project narrative.

Method	AU-ROC	Test	Status
Majority baseline	0.500	0.668	baseline
Paper-style WS (averaged)	0.726 ± 0.030	0.701	replication
Paper-style CNN (ResNet18 → k-NN)	0.762 ± 0.117	0.766	replication
Paper-style CNN + WS fusion (concat)	0.816 ± 0.057	0.792	replication
Scattering covariance ($J=6, L=3$)	0.734 ± 0.062	0.682	extension (tissue-side; no mass gain)
Pooled 2×2 + PCA/RBF-SVM	0.861 ± 0.080	0.792	supporting
Full-patch WS → CNN	0.909 ± 0.053	0.809	spatial-map preservation (key lever)
Region-aware WS (classical, logreg)	0.892 ± 0.041	0.806	extension
Region-aware WS → CNN ★	0.946 ± 0.036	0.899	interpretable lesion-aware instantiation
Improved CNN (square crop, frozen)	0.857 ± 0.059	0.808	comparator
Region-aware late fusion	0.948 ± 0.033	0.883	numerically strongest

Table 5.12 Mass WS-only preprocessing sensitivity (*FINAL Step 2.6*). The same sampled coordinates and labels are used throughout; only the image source changes. AU-ROC and test accuracy are five-seed mean $\pm$ standard deviation. Razali is included as a paper reference. Source: `ws_pixel_source_variant_summary.csv`.

Pixel source	AU-ROC	Test Acc.
Current full preprocessing	0.726 ± 0.030	0.701 ± 0.075
Raw min-max only	0.797 ± 0.021	0.700 ± 0.073
Raw percentile only	0.779 ± 0.045	0.716 ± 0.073
Masked/no CLAHE	0.805 ± 0.045	0.741 ± 0.057
Razali WS-only (paper)	0.800	0.833

Table 5.13 Region-aware WS maps $\rightarrow$ CNN head: current full-preprocessed inputs vs. masked/no-CLAHE inputs. Both rows use the all-four-region tensor (*full+inner+boundary+context*) and the same five-seed protocol. Source: `region_aware_masked_no_clahe_vs_current_comparison.csv`.

Input style	AU-ROC	Test Acc.	F1
Current full preprocessing	0.946 ± 0.036	0.899 ± 0.060	0.923 ± 0.046
Masked/no CLAHE	0.971 ± 0.019	0.925 ± 0.018	0.943 ± 0.016

Table 5.14 Masked/no-CLAHE region ablation for the WS-map CNN head (five-seed protocol). The table is sorted by AU-ROC. Source: `region_ablation_masked_no_clahe_cnnhead_results.csv`.

Region configuration	Type	AU-ROC	Test Acc.	F1
Drop boundary (full+inner+context)	leave-one-out	0.986 ± 0.014	0.900 ± 0.038	0.922 ± 0.032
Drop inner (full+boundary+context)	leave-one-out	0.977 ± 0.009	0.908 ± 0.019	0.930 ± 0.015
Full + context	full+X	0.974 ± 0.032	0.891 ± 0.052	0.911 ± 0.046
Drop context (full+inner+boundary)	leave-one-out	0.973 ± 0.008	0.908 ± 0.019	0.931 ± 0.013
ALL four (full+inner+boundary+context)	all	0.971 ± 0.019	0.925 ± 0.018	0.943 ± 0.016
Full + inner	full+X	0.966 ± 0.031	0.909 ± 0.045	0.930 ± 0.035
Full only	single	0.960 ± 0.030	0.858 ± 0.067	0.884 ± 0.061
Full + boundary	full+X	0.957 ± 0.018	0.900 ± 0.037	0.921 ± 0.031
Inner only	single	0.909 ± 0.036	0.806 ± 0.090	0.846 ± 0.091
Drop full (inner+boundary+context)	leave-one-out	0.908 ± 0.052	0.825 ± 0.052	0.869 ± 0.048
Context only	single	0.902 ± 0.067	0.792 ± 0.104	0.849 ± 0.079
Boundary only	single	0.885 ± 0.054	0.760 ± 0.066	0.798 ± 0.081

Chapter 6

Discussion

6.1 Replication outcomes at a glance

Under the final five-seed protocol (Tables 4.1–4.2) all three paper configurations are reproduced for both tasks. On *tissue*, averaged WS reaches AU-ROC 0.895 (test 0.839), trailing the paper’s reported number by roughly 7–8 pp — the largest single replication gap — while the CNN (0.979) and CNN + WS fusion (0.988) are near-saturated and slightly exceed the paper, which we attribute to a cleaner preprocessing pipeline rather than a methodological advance. On *mass*, averaged WS is weak (0.726 AU-ROC), the CNN-only model reproduces the paper’s reported values to within a few points on the seed-34 metrics (including the overfitting signature in its training curves), and CNN + WS fusion is the strongest strict replication model (0.816 AU-ROC, test 0.792), reproducing the paper’s central claim that handcrafted WS features help a data-starved CNN on small-sample mass classification. The development-narrative figures in Chapter 4 are single seed-34 splits and are superseded by these five-seed means where they differ — most notably the high-variance mass test numbers (e.g. the seed-34 WS-only mass test of 0.52, which rises to a five-seed 0.701).

Beyond strict replication, the supervisor-suggested scattering covariance extension (Section 5.2) provides a stronger WS-branch representation *on tissue*: at $J = 6, L = 3$ it reaches AU-ROC 0.984 / test 0.959 (versus averaged WS’s 0.895/0.839 and the paper’s 0.94/0.914), the strongest adopted WS-branch tissue result in the project. On *mass*, however, it gives no meaningful AU-ROC gain (0.734 versus the 0.726 averaged-WS baseline), so it is adopted as a tissue-side extension only; the mass improvement comes instead from preserving the scattering spatial map and learning over it (Section 5.3), not from a richer scalar descriptor. This is positioned as an improvement to the WS branch, not a paper-replication number, and the comparison to Razali is not apples-to-apples (feature definitions, library, preprocessing, patch counts, and CV scheme all differ).

6.2 Fusion fidelity: what the CNN+WS model does and does not reproduce

The CNN + WS fusion implemented here is a faithful instance of the paper’s Model 5: a feature-level concatenation of the 512-dimensional CNN average-pool vector and the 406-dimensional wavelet-scattering vector (918-d in total), classified by the same Subspace k-NN ensemble as the single-feature models, with no GLCM branch (the paper’s Model 6 adds GLCM but reports no improvement over Model 5). The only unspecified choice we fix is to standardise the concatenated vector on training-fold statistics, which is necessary because the CNN and WS halves occupy very different numerical scales and an un-scaled concatenation would let the larger-magnitude half dominate the distance-based classifier.

On the **tissue** task the fusion reproduces the paper’s qualitative ranking — fusion is the (jointly) strongest model — but not the large fusion *gain* the paper reports. Razali’s CNN + WS rises to 93.5% test from a CNN-only baseline of 77.4%, a +16 pp jump they attribute to handcrafted features compensating an overfitting CNN. In our pipeline the CNN-only tissue model does not overfit (it already reaches $\sim 98\%$ test; Section 4), so fusion has little left to add: our CNN + WS (96.4% test, AU-ROC 0.988 over five seeds) is statistically close to CNN-only (97.9% test, AU-ROC 0.979). The fusion mechanism is thus reproduced only where it is actually needed, and on tissue — with cleaner preprocessing — it is not.

On the **mass** task the paper’s central claim *is* reproduced. Here the CNN genuinely overfits (single-seed CV ≈ 0.99 against an honest held-out accuracy of ~ 0.77), and adding the training-free WS features lifts the honest result: under the leakage-free five-seed protocol, fusion (AU-ROC 0.816, test 0.792) outranks both CNN-only (AU-ROC 0.762) and WS-only (AU-ROC 0.726). This is precisely the regime the paper invokes — wavelet-scattering texture supporting a data-starved CNN — and is the clearest evidence in this project that fusion helps.

One caveat tempers the comparison to the paper’s headline mass numbers. The paper’s reported CNN + WS mass result (91.7% test, AU-ROC 1.00) is a single split on a ~ 25 -patch test set; our *cross-validation* reproduces it ($\approx 98\%$, matching the paper’s 98%), but the honest repeated held-out estimate is lower and noisier. We therefore treat the paper’s CV-led fusion headline as optimistic for very small held-out sets, while affirming its qualitative conclusion that fusion benefits the mass task.

6.3 Sources of the WS-only tissue accuracy gap

The single largest discrepancy is the ~ 7 pp gap on WS-only tissue classification. Three sources are plausible.

- **Contrast enhancement.** The paper uses a density-aware histogram adjustment that the authors describe in earlier work, which adapts to the ACR breast-density category. We use standard CLAHE, which is density-agnostic. The paper’s method is likely to produce features that are more discriminative for the very task being measured (tissue density), because it amplifies density-specific contrast.
- **Library implementation.** Kymatio produces 406-dimensional features (with $L = 5$ orientations, $J = 6$ wavelet scales) whereas the paper’s MATLAB toolbox produces 391 (with $L = 6$, effectively $J = 5$). This is a fundamental library convention difference; we cannot make the two match without re-implementing the scattering transform from scratch.
- **Patch sampling.** Our ACR-guided sampling yields 434 patches versus the paper’s 397, with a slightly different fatty/fibroglandular ratio. ACR guidance prevents ACR-1 images from contributing spurious fibroglandular patches via Otsu thresholding, but it does shift the marginal class distribution slightly.

Of these, the contrast enhancement difference is the most plausibly dominant factor: WS features are essentially a textural descriptor of frequency content, and a density-aware contrast operation literally adjusts the contrast of the very signal WS measures.

Closing the gap by changing the feature representation. A separate observation from the scattering-covariance study (Section 5.2) is that the WS-only gap can also be closed — and substantially exceeded — by enriching the feature representation rather than by changing preprocessing. The adapted scattering covariance at $J = 6, L = 3$ reaches AU-ROC 0.984 on the same patches and same preprocessing pipeline as the original WS, +0.089 above the original WS’s 0.895 (in seed-34 group-CV accuracy terms — a secondary metric — this is 81.2% $\rightarrow$ 96.2%, +15 pp). The interpretation is that the standard Mallat S_0, S_1, S_2 coefficients undersample the cross-scale and cross-orientation information that scattering covariance exposes via its $C_{01}, C_{11}, P_{00}, P_{11}$ statistics. The contrast-enhancement and library-convention issues above remain real, but they no longer dominate the achievable WS-branch performance for this project.

6.4 Lesson learned: image-level vs. patch-level splitting in medical imaging

The most important methodological lesson of the project was uncovered in Stage 2. When multiple patches are extracted per source image, splitting at the patch level allows highly correlated samples to appear in both training and test sets, producing inflated and meaningless test accuracy. In our first CNN run, 98.9% of test patches shared a source mammogram with the training set; the resulting 100% test accuracy was a direct consequence. Image-level splitting (with group-aware cross-validation) is the standard correction in the medical-imaging literature and is now the default choice throughout this codebase. The fact that the paper's wording was ambiguous on this point is worth flagging for anyone attempting a similar replication.

6.5 Mass test-set noise

The held-out mass test set contains only 25 patches across 22 images. A single misclassification moves test accuracy by 4 pp, so a difference of even 5–10 pp between our test estimate and the paper's may not be a real difference in model performance. Cross-validation accuracy is a more stable comparison point at this sample size, and on that metric our CNN-only mass result (95.6%) is close to the paper's (100%) while our WS-only (74.7%) is in fact slightly *better* than the paper's (69%).

6.6 Mass patch count discrepancy with the paper

Our mass dataset contains 116 patches across 107 source mammograms (75 malignant, 41 benign); the paper reports 112 patches (76 malignant, 36 benign), a net difference of +5 benign and −1 malignant. The BI-RADS mapping is not the source of this gap: both pipelines place BI-RADS 2–3 in the benign class and BI-RADS 4–6 in the malignant class (the standard clinical convention). Instead, two distinct properties of our extraction pipeline account for it.

Effect 1 — multi-ROI retention. Eight of our 107 source images contain more than one `Name="Mass"` ROI in the XML annotation (seven contain two, one contains three). Six of those images are BI-RADS-2 benign and contribute seven extra ROIs above one-per-image; two are BI-RADS-5 malignant and contribute two extras. Inspection of the bounding-box centroids reveals that these multi-ROI cases are a mix of two phenomena:

(i) near-duplicate annotations of a single mass — e.g. file 51049107 contains two benign ROIs whose centroids are only 89 pixels apart on bounding boxes 150–240 pixels wide; and (ii) genuinely multifocal disease — e.g. file 22427840 (BI-RADS 5) contains two malignant ROIs whose centroids are 311 pixels apart, consistent with two separate cancer foci in the same breast (multifocality is well documented in high-BI-RADS cases). Our pipeline retains all such ROIs; collapsing to one ROI per image would yield 107 patches (35 benign + 72 malignant) at the cost of dropping genuine multifocal masses.

Effect 2 — “Mass tissue” inclusion criterion. Our *Step 1.2 — Mass ROI Extraction* extracts only ROIs whose XML Name tag is literally Mass. The INbreast dataset additionally contains nine images whose XML carries only Asymmetry or Distortion ROIs and no Mass ROI: six asymmetry-only and three distortion-only. Both the XML Name and the corresponding XLS Mass column agree that these are not masses; nevertheless, eight of these nine images carry BI-RADS scores in the 4a–5 range (i.e. suspicious or confirmed malignancy), so radiologically they may behave like masses even though they have not been annotated as such by the OsiriX reader. The paper does not state whether it accepts these as “mass tissue”.

A plausible reconstruction of the paper’s 112. If the paper takes one Mass ROI per source image (107 patches, 35 benign + 72 malignant) and then adds one BI-RADS-2 distortion image plus four BI-RADS-4/5 asymmetry-or-distortion images, the totals come to $36 + 76 = 112$ — an exact match with their Table 3 counts. This is consistent with two simultaneous editorial choices on the paper’s part: (a) deduplicate multi-ROI images down to one ROI each, and (b) broaden the “mass tissue” definition to include some architectural-distortion and asymmetry findings. The exact subset cannot be identified from the paper text.

Why we retain our extraction. Our criterion is exactly reproducible from the public INbreast XML alone: any reader can re-derive our 116 patches by filtering XML ROIs on `Name == "Mass"`. The paper’s selection, by contrast, requires per-image judgements — which non-Mass mass-like findings to include and which multi-ROI duplicates to collapse — that are not documented and therefore not reproducible without manual review. Keeping all Mass ROIs preserves the clinically important multifocal cases at the small cost of carrying a few near-duplicate annotations; because our train/test split is image-level (Section 4.3), duplicates from the same image never cross the train/test boundary, so the inflated count adds slight redundancy within the training fold but introduces no leakage. Finally, the absolute size of the difference is well below the noise floor of the mass test set ($n_{\text{test}} = 25$): the CNN+WS fusion still reproduces the paper’s recall and F1 to within ~ 1 pp regardless of which counting

convention is used. For these reasons we have not changed the extractor to match the paper’s number exactly.

6.7 Data-quality filter — one entirely-zero mass patch excluded

During the scattering covariance experiment (extension method described in Section 4), a data-quality issue surfaced that had been silently present in every previous mass result. One INbreast annotation — **file_id 50994354**, BI-RADS 4a, labelled malignant, view MG_R_ML_ANON — has a bounding box at $y \in [806, 1098]$, $x \in [2214, 2483]$ on a 3328×2560 preprocessed image. The bbox falls entirely outside the breast mask: *zero of its 78 548 pixels* are inside the segmented breast region (which itself covers over a million pixels, so the breast itself is intact). Step 1.1 zeros all non-breast pixels before saving the preprocessed array, so when Step 1.2 crops the bbox from the preprocessed image the result is a uniformly-zero 224×224 patch.

The root cause appears to be an annotation-or-orientation mismatch between the OsiriX XML and the as-loaded DICOM for this single file: for a right-laterality MLO view the breast itself occupies the left side of the image, so a bbox at $x \in [2214, 2483]$ (very far right of a 2560-wide image) is on the empty chest-wall side. No other mass annotation in INbreast shows this pattern.

Effect. The all-zero patch was in the held-out test split under `random_state=34`, so it directly affected the reported mass test metrics for all four mass-using notebooks (Steps 2.1, 2.2, 2.3, 2.4). It contributes an essentially zero feature vector to whichever classifier it reaches; a k -nearest-neighbour classifier on such a vector lands on whatever class the training set’s nearest non-zero patch happens to be.

Filter applied. A defensive one-line filter was added to every mass-using notebook to exclude patches whose maximum pixel value is zero:

```
mass_df = mass_df[mass_df["patch_npy"].apply(lambda p:
                                              np.load(p).max() > 0)]
```

The filter is dynamic, so it would also catch any future analogous patch, not just this one. The on-disk patches and the upstream `masses/mass_index.csv` are unchanged — only the downstream manifests used by the classifier notebooks were updated. This reduces our mass dataset from 116 patches (75 malignant + 41 benign) to **115 patches (74 malignant + 41 benign)**.

Numerical impact. On the post-filter mass results (24-patch test set), every method’s test accuracy improves by 1–3 pp, since the all-zero patch was a guaranteed-wrong prediction for most methods. CV mean and CV std are essentially unchanged because the patch was in the test split, not in any training fold. The qualitative ranking of methods is preserved. The numerical mass figures reported earlier in this document — the WS-only and CNN+WS rows of Table 4.2, the $n_{\text{test}} = 25$ in Section 4 Stage 5a/b, and the mass-specific numbers cited in Sections 4 and 6 — were computed before the filter was applied. Re-running Steps 2.1, 2.2, and 2.3 with the now-filtered mass dataset would shift each test number by $\sim 1\text{--}3$ pp upward without changing the qualitative picture; we have not re-run them here, but the filter has been added to those notebooks so any future execution will automatically use the 115-patch dataset.

A wider Step 1.2 improvement — adding a bbox-vs-breast-mask overlap check at extraction time, or cropping from the polygon outline rather than the rectangular bbox — is deferred as a longer-term cleanup; the one-line filter is sufficient for the current scope.

6.8 Annotation source audit — inter-source disagreements within INbreast

INbreast ships with several overlapping annotation sources. Two of these inform the pipeline directly: the lesion ROIs (used to extract mass patches) and the pectoral muscle outlines (used in preprocessing). For each, two independent versions exist in the public release, and they do not perfectly agree. The full audit, including visual side-by-side comparisons, is in the project notebook *Step 0.3 — Investigate Annotation Sources*; the summary below records the conclusions reached there.

Lesion annotations — XML ROIs vs. XLS Mass column

The presence of a mass in each case is recorded independently in two places:

- The `INbreast.xls` metadata file has a `Mass` column. The radiologist ticked this with X where a mass was reported. This is a per-file binary flag with no spatial information; **108** files are flagged.
- The OsiriX XML files in `AllXML/` contain ROI polygons. A subset of ROIs carry the tag `Name="Mass"`, with pixel-coordinate vertex lists drawn by the (separate) annotator who created the XML files; **107** files have at least one `Mass` ROI.

Of these, **106** files appear in both sets. The 3 disagreements are:

- 22614097 (BI-RADS 2, L CC) and 22614150 (BI-RADS 2, L MLO): the XLS marks a mass with X, but the XML annotator drew only calcification ROIs (3 and 4 calcifications respectively), no mass polygon. These are the two XLS-only files. Both are also flagged with `Micros='X'` in the XLS; the most plausible reading is that the case carried a clinically-benign mass that the XML annotator either considered too small to outline or merged into the calcification cluster.
- 24065584 (BI-RADS 4c, L CC, ACR 4): the XLS lesion columns are all empty, but the XML annotator drew an unambiguous large Mass polygon. This is the one XML-only file. The XML annotator's outline is consistent with the BI-RADS 4c (highly suspicious for malignancy) classification; the XLS row appears to have simply been left blank.

The arithmetic checks out: $108_{\text{XLS}} - 2_{\text{XLS-only}} + 1_{\text{XML-only}} = 107_{\text{XML}}$. The mass-extraction pipeline (Step 1.2) takes ROIs from the XML because it requires polygon coordinates to crop patches from; the XLS Mass column alone provides no spatial information. The choice is therefore forced by what the pipeline needs, but on the merits it works in our favour: the two XLS-only files we miss are both BI-RADS 2 (low-acuity benign), while the one XML-only file we gain is BI-RADS 4c on a dense ACR-4 breast — exactly the kind of high-acuity malignant case the mass classifier benefits from seeing. The net 107-vs-108 difference is a real inter-annotator inconsistency within INbreast itself, not an artefact of our extraction.

Pectoral muscle masks — GitHub LabelMe-JSON vs. Kaggle Apple-plist XML

The preprocessing pipeline (Section 3.2) zeroes out the pectoral muscle on MLO/ML views using ground-truth polygon masks. Two public sources of these masks exist:

- **GitHub source** — LabelMe-format JSON files by Aliniya [19], covering **202** MLO views. This is the source used by the current pipeline.
- **Kaggle source** — Apple-plist XML files distributed with the original INbreast 1.0 release [7] (INESC Porto), covering **201** MLO views.

The two sources agree on the `file_ids` of **199** cases, with 3 GitHub-only (20587492, 20587518, 50994273) and 2 Kaggle-only (24055024, 53582737). The GitHub-only 50994273 is an unusual case: it is a CC view (no pectoral muscle should be visible in principle), and

the GitHub annotator nonetheless drew a polygon for a small sliver of muscle in the corner that the original Porto annotators did not annotate.

On the 199 files both sources annotate, the two masks are visibly different in style:

- The GitHub masks are coarse **triangular regional outlines** (typically 3–8 polygon vertices), filling the upper-corner area where the pectoral muscle is.
- The Kaggle masks are fine **muscle-boundary outlines** (typically 20–40 vertices), hugging the actual muscle tissue.

For our pipeline’s purpose — zeroing the muscle region before breast-mask + CLAHE in Step 1.1 — either source works. The GitHub source removes the entire upper-corner region (muscle plus some surrounding background); the Kaggle source removes only the muscle tissue itself. The downstream impact on classification is expected to be small because (i) the Otsu breast mask removes non-breast pixels regardless of which pectoral mask is used, and (ii) the patch sampler in Step 1.3 enforces a minimum target-fraction (`min_target_frac`), which suppresses patches that straddle the muscle edge. We retain the GitHub source as the default for two reasons: it is already wired into Step 1.1, and its slightly more conservative coverage removes a small amount of additional non-breast pixels that the Kaggle outlines leave behind.

6.9 Overfitting signatures

The supervisor’s concern about CNN overfitting is borne out by the per-epoch training/validation curves logged by the Stage 4 and Stage 5b notebooks. Training accuracy saturates at 100% within the first 10 epochs in both tasks. For the tissue task, validation accuracy then stays close to training accuracy, which is why the overfit gap is small. For the mass task, validation accuracy oscillates in the 60–90% range, mirroring the paper’s reported large CV–test gap and supporting their argument that fusion with handcrafted features is necessary for small-sample mass classification.

6.10 What the extension shows

The extension experiments converge on a single message: for the mass task, *preserving spatial and lesion structure in the wavelet-scattering representation matters more than changing the scalar features or the classifier*. The screening of Section 5.1 shows that normalisation, per-path statistics, coarse pooling, feature selection, and even the richer

scattering-covariance descriptor move mass AU-ROC only modestly and within the across-seed noise — clustered near the 0.726 averaged-WS baseline and rising to ~ 0.86 for the best hand-pooled variant. The one method that breaks out of that band is the full-patch WS $\rightarrow$ CNN head (0.909), which keeps the $J = 3$ spatial maps and learns over them rather than averaging them away. Conditioning those maps on lesion morphology (region-aware WS) lifts the *classical* (logistic-regression) model from 0.801 to 0.892; with a learned CNN head the region-aware maps reach 0.946 (group-aware CV 0.938 ± 0.007), above the whole-patch map+CNN (0.909). The region ablation shows the all-four-region tensor is the strongest complete configuration, although neighbouring strong configurations overlap within seed noise. The operative lever is still preserving the spatial map and learning over it, with the region split retained as a principled, interpretable representation that now also gives the strongest complete region-aware result.

The contrast with tissue is instructive. Tissue density is a texture problem that a richer *scalar* descriptor — scattering covariance — already solves (0.984 AU-ROC), with no need to retain spatial layout. Mass is a structured-object problem, where *where* the discriminative signal sits (core versus margin versus surrounding context) is decisive, so the scalar route plateaus and the spatial route succeeds. Region-aware preservation is therefore the project’s main methodological contribution on the mass side. The CNN and fusion methods support that conclusion rather than replace it: the region-aware late fusion is numerically strongest (0.948), but it is a decision-level combination of two trained models, not a new representation, and its margin over the region-aware WS $\rightarrow$ CNN alone is within seed noise — so it is reported as the best *number*, while the region-aware representation is the substantive *result*.

6.11 Reproducibility: determinism and holdout-vs-CV stability

Three reproducibility issues shaped the final evaluation protocol (Section 3.9).

Small- n instability and seed sensitivity. The mass held-out set is only ~ 24 patches, so a single split can swing by 10 pp or more between seeds (the seed-34 WS-only mass test of 0.52 versus a five-seed mean of 0.701 is the clearest example). Every headline mass method is therefore reported as a five-seed mean $\pm$ standard deviation over image-level splits, and the residual bands (e.g. ± 0.036 for the region-aware WS $\rightarrow$ CNN) are an honest reflection of the small sample rather than a defect of the method.

Determinism of learned models. For the learned WS-map CNN, Python/NumPy/PyTorch seeds and cuDNN-deterministic flags are pinned, but the model is still trained on only ~ 90

patches. The mitigation is to average five seeds and corroborate with a tight group-aware cross-validation (0.938 ± 0.007), which is far less seed-sensitive than any single held-out split.

Optimistic CV for learned features. Cross-validation computed on *learned* CNN or fusion features is optimistic, because the feature extractor has effectively already seen the inner-validation fold. The CNN-only and fusion CV figures of ~ 0.99 on mass (marked “(opt.)” in Table 4.2) are not honest out-of-sample estimates, whereas their five-seed held-out AU-ROCs (0.762 and 0.816) are. For this reason the project leads with AU-ROC under the repeated image-level holdout as the headline mass metric, and uses group-aware CV only as a stability cross-check — and, for the classical WS methods where no feature is learned, as a like-for-like paper-comparison metric.

6.12 Limitations

- **Small mass sample.** The mass dataset is only 115 patches (~ 24 held out per split). The final protocol averages five image-level seeds and reports standard deviations to expose this, but the absolute sample remains small, so the mass AU-ROC bands stay wide; this is the dominant limitation on the mass side.
- **Single dataset.** Only INbreast is used. Cross-dataset generalisation to e.g. DDSM or MIAS has not been tested.
- **No augmentation.** Following the paper, no augmentation is used. Augmentation is the most obvious lever to mitigate the observed CNN overfitting.
- **StandardScaler.** The paper does not mention applying feature standardisation before the Subspace k-NN; we do (since k-NN is distance-based). This may help our WS results slightly, but the overall qualitative picture (e.g. WS-tissue trailing the paper) is unchanged.
- **GLCM branch excluded.** On supervisor advice, the GLCM branch and the CNN+GLCM/CNN+WS+ fusion configurations (Models 3, 4, 6 in the paper’s Table 6) are out of scope. The paper itself reports that adding GLCM to CNN+WS yields no change on accuracy, precision, recall or F1 and only a $+0.01$ AU-ROC gain at the cost of additional training time, so the practical impact of this exclusion is small.
- **Mass AU-ROC volatility.** Even averaged over five seeds, mass AU-ROC carries a visible band (replication fusion 0.816 ± 0.057 ; region-aware WS $\rightarrow$ CNN $0.946 \pm$

0.036), because the held-out set is tiny and the probability ranking is coarse. The paper’s reported 1.00 fusion AU-ROC is a single ~ 25 -patch split and is not a stable target; a larger external mass set would be needed for a tight estimate.

- **Scale not fully optimised for mass.** The scattering-covariance J/L tuning was carried out on the *tissue* task (Section 5.2). On mass we ran only an L -control at $J = 6$ (which showed no gain) and adopted $J = 3$ maps for the whole-patch and region-aware models, consistent with Bruna and Mallat’s [4] observation that an object-like task favours a *smaller* invariance scale that retains spatial structure. A full mass-specific covariance $J \times L$ grid was not run; the region-aware results, however, indicate the larger lever for mass is preserving spatial structure rather than re-tuning the invariance scale of a scalar descriptor.

6.13 Future work

Beyond what is in this report, work that would further strengthen the project includes: replacing CLAHE with the paper’s density-aware histogram adjustment to test whether this closes the residual WS-tissue gap in the original Mallat representation; introducing light augmentation to reduce CNN overfitting (a departure from the paper but well-motivated by the training/validation curves logged for the Section 4 experiments); **external validation** on a second mammography dataset (e.g. CBIS-DDSM or MIAS) to test whether the region-aware mass improvement generalises beyond INbreast; reproducing the CNN+WS fusion — and the region-aware late fusion — on top of the scattering-covariance vector rather than the standard S_0, S_1, S_2 vector; running a *full* mass-specific scattering-covariance $J \times L$ grid (only an L -control at $J = 6$ was run here), since Bruna and Mallat [4] indicate an object-like task may favour a smaller invariance scale; and investigating the effect of BI-RADS 3 placement (benign vs malignant) on the mass numbers. Multi-seed averaging, previously identified as the single most impactful next step, is now the default evaluation protocol throughout this report rather than outstanding future work.

Chapter 7

Conclusion

This report has presented a Python replication of the wavelet scattering, CNN, and CNN + WS fusion branches of Razali et al. [5], evaluated on both the fatty versus fibroglandular tissue task and the benign versus malignant mass task, together with two extensions to the wavelet-scattering branch: a supervisor-suggested scattering covariance representation [8] for the tissue task, and a region-aware wavelet-scattering decomposition developed for the mass task. The work proceeded iteratively: an early CNN run on the tissue task surfaced a data-leakage issue that prompted a re-examination of the train–test split methodology for both branches; only after that correction was the mass-classification task added, followed by the fusion model and finally the scattering covariance extension. The chronological development log is in Section 4; Tables 4.1 and 4.2 compare every replicated configuration against the paper’s reported benchmarks.

Under the final five-seed image-level protocol (Tables 4.1–4.2), all three paper configurations reproduce for both tasks. On *tissue*, WS-only reaches AU-ROC 0.895 ± 0.070 (test 0.839 ± 0.071), trailing the paper by roughly 7–8 pp — the largest replication gap, attributable to standard CLAHE versus the paper’s density-aware contrast and to scattering-library conventions — while the CNN-only (0.979 ± 0.010 AU-ROC, test 0.979 ± 0.010) and CNN + WS fusion (0.988 ± 0.007 , test 0.964 ± 0.030) are near-saturated and slightly exceed the paper, reflecting a cleaner preprocessing pipeline rather than a methodological change. On *mass*, WS-only is weak (AU-ROC 0.726 ± 0.030 , test 0.701 ± 0.075), the CNN-only model reproduces the paper’s reported values to within a few points on the seed-34 metrics (including the overfitting signature in its training curves), and CNN + WS fusion is the strongest strict replication model (AU-ROC 0.816 ± 0.057 , test 0.792 ± 0.058) — above both CNN-only (0.762 ± 0.117) and WS-only (0.726) — directly reproducing the paper’s central narrative that handcrafted scattering features compensate for small-sample CNN overfitting.

The scattering covariance extension (Section 5.2) was tested against the original WS baseline, a direct usage of the Cheng package, and the paper’s published WS-only number. The final adopted WS-branch representation is the adapted scattering covariance configuration with $J = 6, L = 3$ on the `for_synthesis_iso` vector (580 scalar coefficients per patch), selected by a small J/L grid search ranked by 10-fold group-aware CV mean on the training partition. This configuration reaches AU-ROC 0.984 ± 0.010 / test 0.959 ± 0.013 on tissue (versus averaged WS’s $0.895/0.839$), the strongest adopted WS-branch tissue result in the project. On *mass*, however, it gives no meaningful AU-ROC gain (J6/L3 scattering covariance 0.734 ± 0.062 versus averaged WS 0.726 ± 0.030), so it is adopted as a tissue-side extension only; the mass improvement comes instead from preserving the scattering spatial map and learning over it (Section 5.3). The scattering covariance representation respects the supervisor’s no-flattening constraint by construction: its outputs are scalar coefficients indexed by scale and orientation tuples, never a stack of flattened spatial maps. The original Mallat-style WS implementation is retained as the faithful paper-replication reference; the Direct Cheng usage at $L = 4$ is retained as an independent sensitivity check confirming the result is not specific to the $L = 3$ choice. The CNN+WS fusion experiments (Section 4.6) used the original S_0, S_1, S_2 WS features for direct comparability with the paper; reproducing fusion on top of the scattering covariance features is identified as natural further work.

The GLCM branch (Models 3, 4, 6 in the paper’s Table 6) was excluded on supervisor advice; the paper itself shows GLCM contributes negligible improvement when combined with WS, so the practical loss is small. A small data-quality issue was uncovered during the scattering covariance experiment: one mass annotation (file_id 50994354, Section 6.7) had a bounding box outside the breast mask, producing a uniformly-zero patch. A defensive filter was added to every mass-using notebook; the test set went from 25 to 24 mass patches with no qualitative change to method rankings.

Beyond the tissue-side scattering covariance, the project’s main *mass*-side extension is to preserve the wavelet-scattering spatial map and learn over it with a small CNN (Section 5.3). Because averaged WS discards *where* lesion structure sits, the maps are retained at $J = 3$ rather than collapsed to a scalar and a small WS-map CNN learns over them; in the lesion-relative *region-aware* instantiation each lesion is additionally decomposed into core/inner, boundary, whole-lesion, and surrounding-context regions before scattering. This recovers most of the lost ground: a whole-patch $J = 3$ -map CNN reaches AU-ROC 0.909 ± 0.053 , the region-aware classical (logistic-regression) model reaches 0.892 ± 0.041 , and the region-aware WS→CNN head reaches 0.946 ± 0.036 on the five-seed holdout (corroborated by a tight group-aware cross-validation of 0.938 ± 0.007) — all well above both the 0.726 averaged-WS baseline and the 0.816 paper-style fusion. A region ablation shows the all-four-

region tensor is the strongest complete configuration, while neighbouring strong variants overlap within seed noise, so the region split is retained as a principled, interpretable lesion representation and as the current strongest complete region-aware result. A decision-level *late fusion* of the region-aware head with a frozen-ResNet improved-CNN comparator (0.857 ± 0.059) is the numerically strongest configuration (0.948 ± 0.033 AU-ROC, test 0.883), but it is an ensemble of two models rather than a new representation, and its margin over the region-aware head alone is within seed noise; it is therefore reported as the best *number*, not the central contribution.

The methodological takeaway is twofold. For the *tissue* (texture) task, a richer *scalar* wavelet-scattering descriptor — scattering covariance — is effective and sufficient. For the *mass* task, by contrast, *preserving the spatial structure* of the scattering representation — retaining its lower-scale maps and learning over them with a CNN — matters more than any change of scalar feature or classifier; conditioning those maps on lesion morphology (core, margin, surrounding context) makes that spatial representation interpretable and, in the current seed-99 results, numerically strongest among the complete region-aware configurations. This is the project’s principal finding beyond replication.

These conclusions are bounded by the usual caveats. The INbreast mass set is small (115 patches, ~ 24 held out per split), so every mass estimate carries an across-seed band and is reported as a five-seed mean $\pm$ standard deviation under strict image-level (group-aware) splitting to avoid the patch-level leakage diagnosed early in the project; no data augmentation is used, following the paper; and only a single dataset is evaluated, leaving cross-dataset generalisation to future work. The region-aware WS $\rightarrow$ CNN result is reported as a five-seed 0.946 holdout, corroborated by the 0.938 group-aware cross-validation.

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Appendix A

Software and Dependencies

All code is implemented in Python. The key libraries used are:

- **pydicom** [20] — DICOM file reading.
- **OpenCV** [23] — image processing (thresholding, morphology, CLAHE, resizing).
- **Kymatio** [24] — wavelet scattering transform computation.
- **scikit-learn** [16] — k-NN classifier, cross-validation, StandardScaler, evaluation metrics.
- **NumPy, Pandas, Matplotlib** — numerical computation, data handling, and visualisation.

The vendored `scattering_transform` package (Cheng & Ménard) is imported locally for the scattering-covariance experiments and does not require installation. Exploratory, negative-control, and superseded methods that were tried but not adopted are catalogued separately in Appendix B.

Appendix B

Exploratory and Negative-Control Methods

Several wavelet-scattering and CNN/fusion variants were explored during the project but were *not* selected as final methods. They are preserved with full provenance in the appendix notebook *FINAL Step 2.5 — Extension Methods Appendix* and summarised in Table B.1. They document the breadth of the search and act as negative controls supporting the concise discussion in Section 5.5; none was adopted because it was unstable, superseded, computationally heavy, or did not beat a simpler baseline.

Table B.1 Appendix-level extension methods preserved in *FINAL Step 2.5* — provenance and negative-control results, not final methods.

Method	Purpose	Outcome	Why appendix-level
Full-map flattening / flatten+PCA	test whether flattening the scattering map helps	no gain over averaged or structured WS	against the no-flattening guidance; superseded
Tensor-distance WS classifiers	classify by inter-tensor distance instead of feature vectors	did not beat the spatial-mean baseline	heavier to compute, no benefit
WS J -sweeps / granularity tests	find the scale/grid that best retains structure	pointed to lower- J maps with a learned head	diagnostic; informed the main method
Extra CNN crop / classifier variants	sweep crop size and frozen-feature heads	confirmed square crop + frozen ResNet + logreg as comparator	screening; clean comparator owned by Section 5.3
Extra fusion variants	alternative early / late fusion heads	head-sensitive; no robust whole-patch gain	screening; headline fusion owned by Chapter 5

Appendix C

Declaration of Use of Autogeneration Tools

Claude (Anthropic) assisted the author in developing and debugging code for the preprocessing and classification pipeline. All experimental design decisions and analysis are the author's own. A detailed declaration of the use of generative tools is provided in the project repository.

